DNS LOG ANALYSIS

SOC Investigation Report

DNS Security Investigation with Splunk SIEM

Student name	[Your full name]
Batch	[Your batch]
Project level	SOC Analyst / L1
Primary tool	Splunk Enterprise 10.6.0.5
Dataset	Zeek `dns.log`, 427,935 DNS events, 16 to 17 March 2012
Report date	4 October 2026
Final assessment	Suspicious and requires validation (host 192.168.202.79)

Learn • Search • Investigate • Correlate • Report

Authorised training dataset analysed in an isolated lab environment.

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Conventions: all times are UTC unless marked IST (Splunk displays IST, UTC + 5:30). "Figure N" refers to the screenshots in this report; "SPL N" refers to the searches listed in Appendix A.

1. Executive Summary

This report documents a DNS threat-hunting investigation performed in Splunk Enterprise on a Zeek `dns.log` containing **427,935 DNS events** captured between **16 March 2012 12:30 UTC** and **17 March 2012 20:59 UTC**. The investigation followed the four phases of the project brief: data validation, baseline analysis, advanced threat hunting and threat-intelligence validation.

The baseline is dominated by normal background activity: Windows IPv6 (Teredo) tunnelling, Google, Apple and Symantec services, software-update checks and NetBIOS name resolution. The busiest client, `10.10.117.210` (75,943 queries), repeats only about 250 domains and is assessed as expected behaviour pending asset confirmation.

The most significant finding is internal host **192.168.202.79**, which shows a combination of behaviours that no other host shows:

- **437 of the 440 zone-transfer (AXFR) requests** in the dataset, covering 7 `hec.net` zones across 10 DNS servers. **415 requests received NOERROR responses** whose answers contained internal hostnames and IP addresses.
- **3,703 reverse (PTR) lookups** spread evenly across nine subnets (192.168.21.0 to 28.0 and 192.168.229.0), **2,620 unique domains** queried and **2,557 unique failed names**.
- DNS blocklist (DNSBL) lookups from 192.168.202.79 and 192.168.202.108, and `version.bind` software-fingerprinting queries from 192.168.202.102 and 192.168.202.115.
- The domain `rootshell-security.net` was requested 770 times by seven clients, all answered NXDOMAIN. Threat intelligence is mixed (VirusTotal 4 of 91 vendors; URLhaus no entry) and is treated as inconclusive.

> **Overall assessment: SUSPICIOUS AND REQUIRES VALIDATION**
>
> The behaviour of 192.168.202.79 matches network reconnaissance of the `hec.net` environment. DNS logs cannot show intent, the tool used, or how much data was transferred, and an approved vulnerability scan or penetration test would produce the same pattern. The activity is therefore **not** classified as malicious.
>
> **Priority actions:** (1) confirm whether the activity was authorised and identify the owner of 192.168.202.79; (2) confirm what was transferred using firewall/`conn.log` and DNS-server logs; (3) restrict AXFR on the DNS servers to approved secondary servers only.

2. Environment and Investigation Data Source

2.1 Environment summary

Item	Detail
Tool	Splunk Enterprise 10.6.0.5 (local lab instance)
Data source	Zeek `dns.log` (tab-separated, no header row, about 62 MB). The original file was kept unchanged.
Index / sourcetype	`dns_lab` / `dns_sample`
Events	427,935
Time range (UTC)	16 Mar 2012 12:30:05 to 17 Mar 2012 20:59:51
Time display	Splunk displays times in the user time zone (IST, UTC + 5:30). This report states UTC unless a value is marked IST. Subtract 5 h 30 min from Splunk's displayed time to obtain UTC.
Field extraction	Search-time `rex` macro named `dnsf` (see Appendix B). Fields follow the Zeek DNS log layout.
Authorisation	Authorised training dataset analysed inside the lab environment only.

2.2 Data ingestion and validation

The file was uploaded with **Settings → Add Data → Upload**. Zeek records begin with an epoch timestamp, and the events are from 2012, which is older than Splunk's default acceptance window. Splunk therefore stamped the first preview with the upload time. The sourcetype `dns_sample` was saved with the settings below so that Splunk reads the real event time:

Setting	Value	Purpose
`TIME_PREFIX`	`^`	Timestamp is at the start of each line
`TIME_FORMAT`	`%s.%6N`	Epoch seconds with microseconds
`MAX_TIMESTAMP_LOOKAHEAD`	`17`	Length of the epoch timestamp
`TZ`	`UTC`	Zeek logs epoch time (UTC)
`MAX_DAYS_AGO`	`10951`	Accept timestamps older than the default limit (2012 data)
`DATETIME_CONFIG`	`/etc/datetime.xml`	Enable timestamp detection
`SHOULD_LINEMERGE`	`false`	One event per line

Validation. Figure 1 shows that events are searchable, the raw records are visible, `host`, `source` and `sourcetype` are correct, and the timestamps are in March 2012 (Splunk lists the newest events first; 3/18/12 2:29 AM IST equals 17 Mar 20:59 UTC). The total event count and time range are confirmed in Figure 4.

SPL 1: Validation search

```
index=dns_lab sourcetype=dns_sample | head 20
```

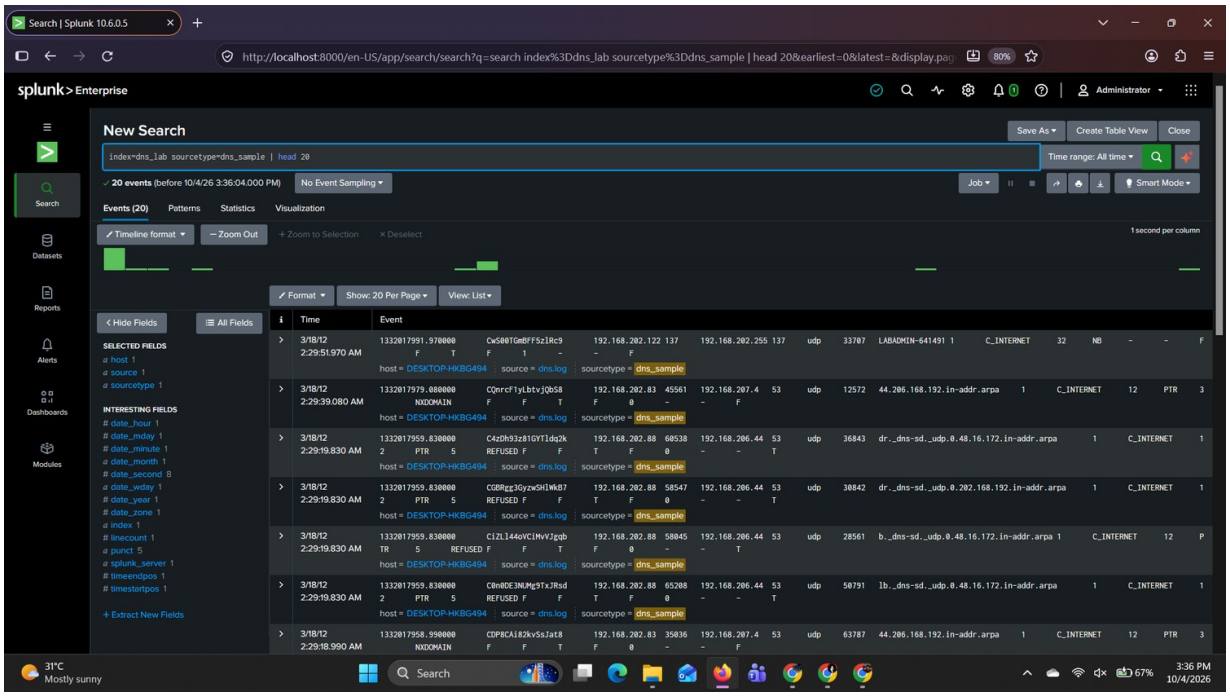


Figure 1: Validation search showing 2012 timestamps and raw DNS records

3. Methodology

The investigation followed the project phases and the SOC rules of engagement (evidence first, reproducibility, context matters):

1. **Phase 1: Understand the data.** Validated ingestion and discovered the fields present (Section 4.1).
2. **Phase 2: Baseline.** Completed Tasks 01 to 10: totals, top domains and clients, query types, response codes, activity over time, client/domain pairs, rare domains, NXDOMAIN and keyword searches (Section 4.2).
3. **Phase 3: Advanced hunting.** Compared query volume with unique domains, domains with many clients, repeated NXDOMAIN by client, query bursts and a domain pivot (Section 5).
4. **Phase 4: Threat intelligence.** Checked a domain of interest against VirusTotal and URLhaus and documented the limitations (Section 6).
5. **Case development.** The strongest lead (192.168.202.79) was pivoted by query type, response code, frequency and related clients (Sections 5.6 to 5.9 and 9).

Facts and assumptions. Throughout this report, *facts* are values returned by Splunk and are tied to a figure and an SPL search. *Assessments* are interpretations and are labelled as such. Popularity, rarity or a keyword alone is never treated as proof of malicious activity.

Reading the log. A `-` in `response_code` means Zeek logged the query but did not record a matching reply. It is not a DNS error code.

4. DNS Baseline

4.1 Phase 1: Field discovery

Field discovery shows which fields exist before any analysis. Figure 2 shows the full summary. It also lists fields that Splunk adds automatically from the timestamp (`date_hour`, `date_mday`, `date_minute`), which are not DNS fields. `fieldsummary` examines the first 400,000 events only, so it is used here for field discovery and not for totals.

SPL 2: Field summary

```
index=dns_lab sourcetype=dns_sample | `dnsf` | fieldsummary | sort - count
```

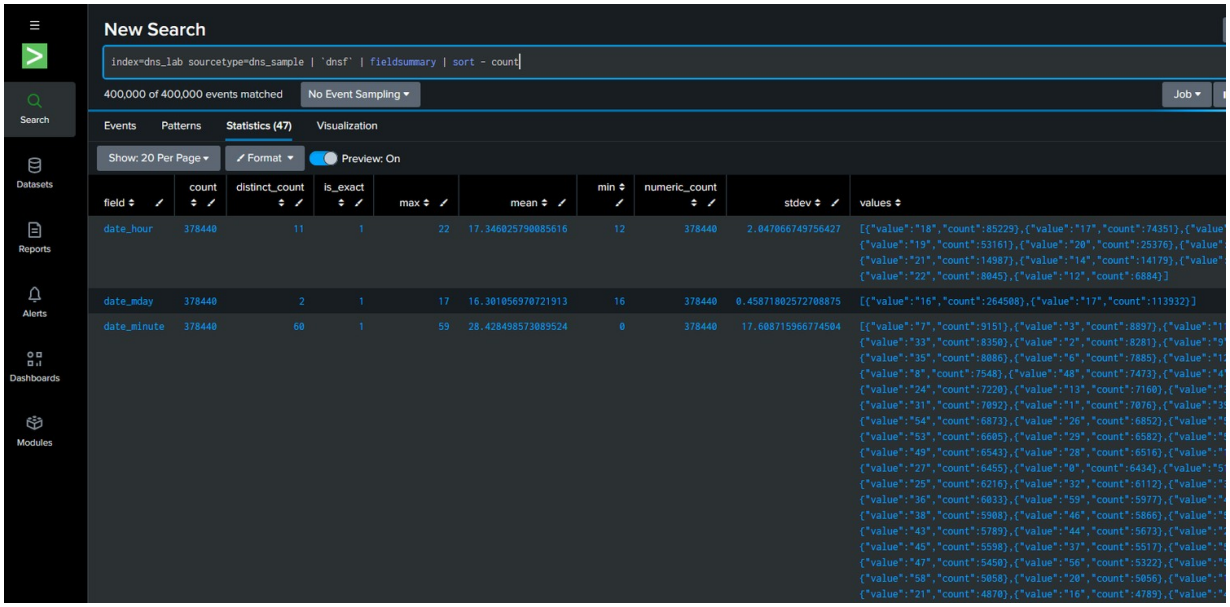


Figure 2: Field summary (first 400,000 events), including Splunk's automatic date fields

Figure 3 narrows the result to the DNS fields used in the investigation. All ten fields are present on all 427,935 events, which confirms that the extraction macro works on every record.

SPL 3: DNS fields only

```
index=dns_lab sourcetype=dns_sample | `dnsf` | fieldsummary | search field IN
("src_ip","dest_ip","fqdn","query_type","response_code","src_port","dest_port","proto","uid"
,"ts") | table field count distinct_count | sort - count
```

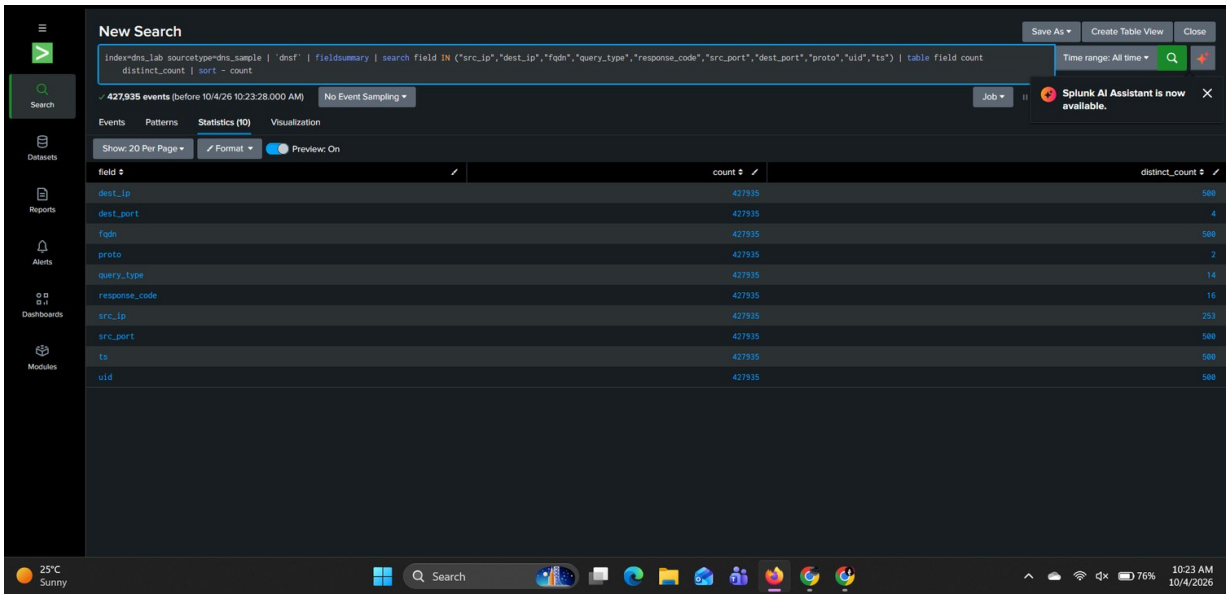


Figure 3: DNS fields extracted on all 427,935 events

Field	Meaning	Investigation use
`src_ip`	Client making the DNS request (253 distinct clients)	Find the most active or unusual clients
`dest_ip`	DNS server or destination of the query	Identify resolver infrastructure and targeted servers
`fqdn`	Domain name requested	Investigate suspicious or rare destinations
`query_type`	Record type (A, AAAA, PTR, TXT, AXFR, NB ...); 14 types	Detect unusual query patterns
`response_code`	Result of the query (NOERROR, NXDOMAIN, REFUSED, `-` ...); 16 values	Find failure-heavy behaviour
`dest_port` / `proto`	Service port (53, 137, 5353, 5355) and transport (udp, tcp)	Separate DNS from NetBIOS, mDNS and LLMNR; identify TCP zone transfers
`ts`, `uid`, `src_port`	Event time (epoch), Zeek connection ID, client port	Timelines and correlation

Note: `fieldsummary` caps distinct-value counts at 500, so `dest_ip`, `fqdn`, `ts`, `uid` and `src_port` show 500 in Figure 3. The true numbers are higher (for example, 5,178 unique domains calculated from the same log).

4.2 Phase 2: Core investigation tasks

Task 01: Baseline the data source

SPL 4

```
index=dns_lab sourcetype=dns_sample | stats count min(_time) as first max(_time) as last |
convert ctime(first) ctime(last)
```

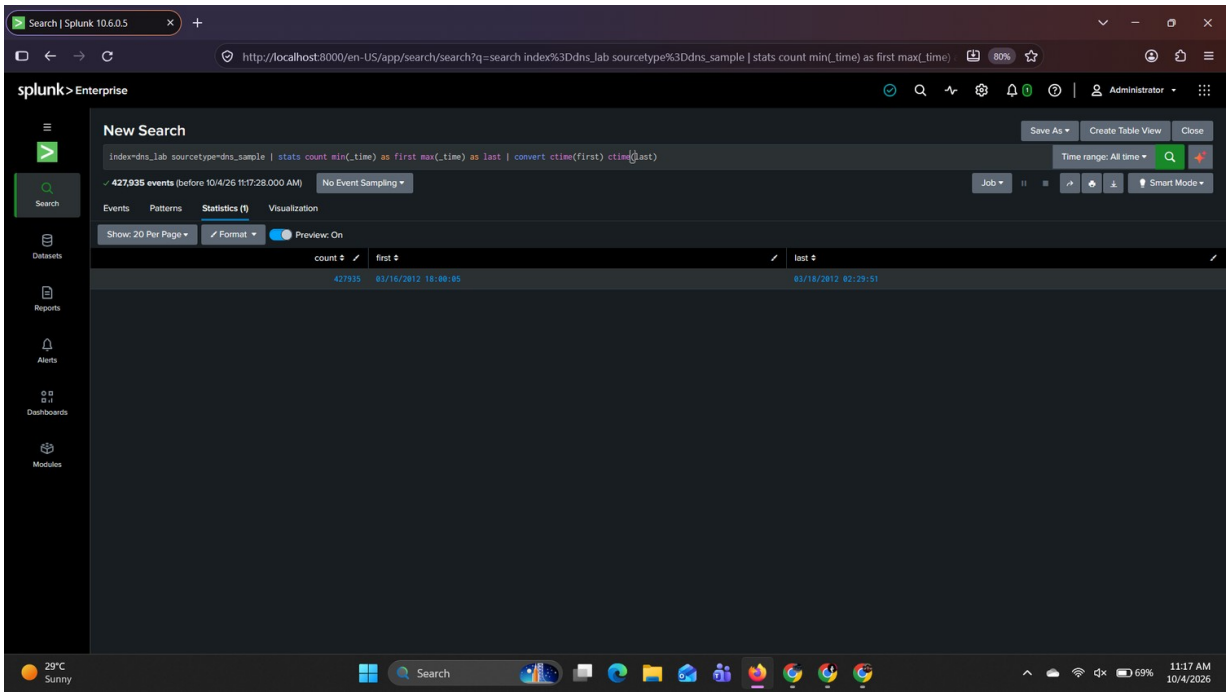


Figure 4: Total events and time range (Splunk displays IST)

Result: 427,935 events from **16 Mar 2012 12:30:05 UTC** to **17 Mar 2012 20:59:51 UTC**. Splunk displays 03/16/2012 18:00:05 and 03/18/2012 02:29:51 because the user time zone is IST (UTC + 5:30).

Task 02: Top queried domains

SPL 5

```
index=dns_lab sourcetype=dns_sample | `dnsf` | top limit=20 fqdn
```

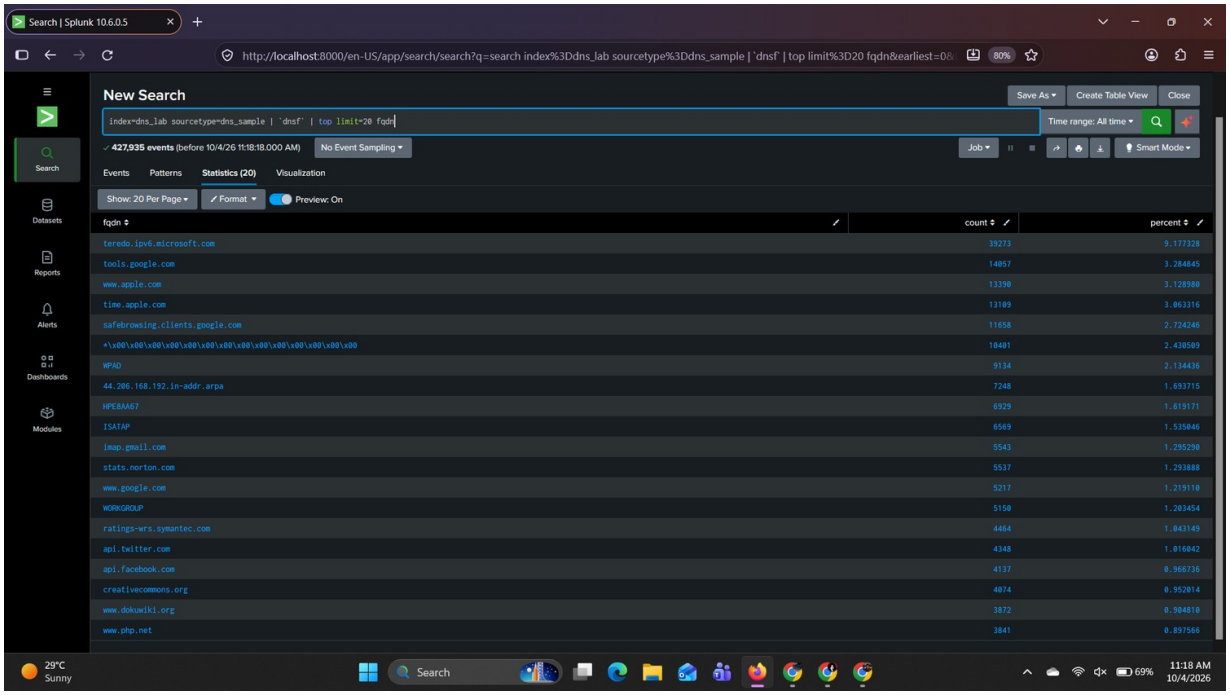


Figure 5: Top 20 queried domains

#	Domain	Queries	Typical purpose
1	`teredo.ipv6.microsoft.com`	39,273	Windows IPv6 (Teredo) tunnelling
2	`tools.google.com`	14,057	Google toolbar / update checks

#	Domain	Queries	Typical purpose
3	`www.apple.com`	13,390	Apple connectivity and updates
4	`time.apple.com`	13,109	Time synchronisation
5	`safebrowsing.clients.google.com`	11,658	Browser safe-browsing checks
6	NetBIOS wildcard name (`*\x00...`)	10,401	NetBIOS name service
7	`WPAD`	9,134	Proxy auto-discovery (local name)
8	`44.206.168.192.in-addr.arpa`	7,248	Repeated reverse lookup of one internal address
9	`HPE8AA67`	6,929	Local NetBIOS name
10	`ISATAP`	6,569	IPv6 transition (local name)

Interpretation: The most frequent domains are well-known services performing routine background work, plus local NetBIOS names. High frequency alone is **not** enough to call a domain malicious, because automated software legitimately repeats the same lookups thousands of times. Frequency is only a prompt to check context.

Task 03: Top DNS clients

SPL 6

```
index=dns_lab sourcetype=dns_sample | `dnsf` | top limit=20 src_ip
```

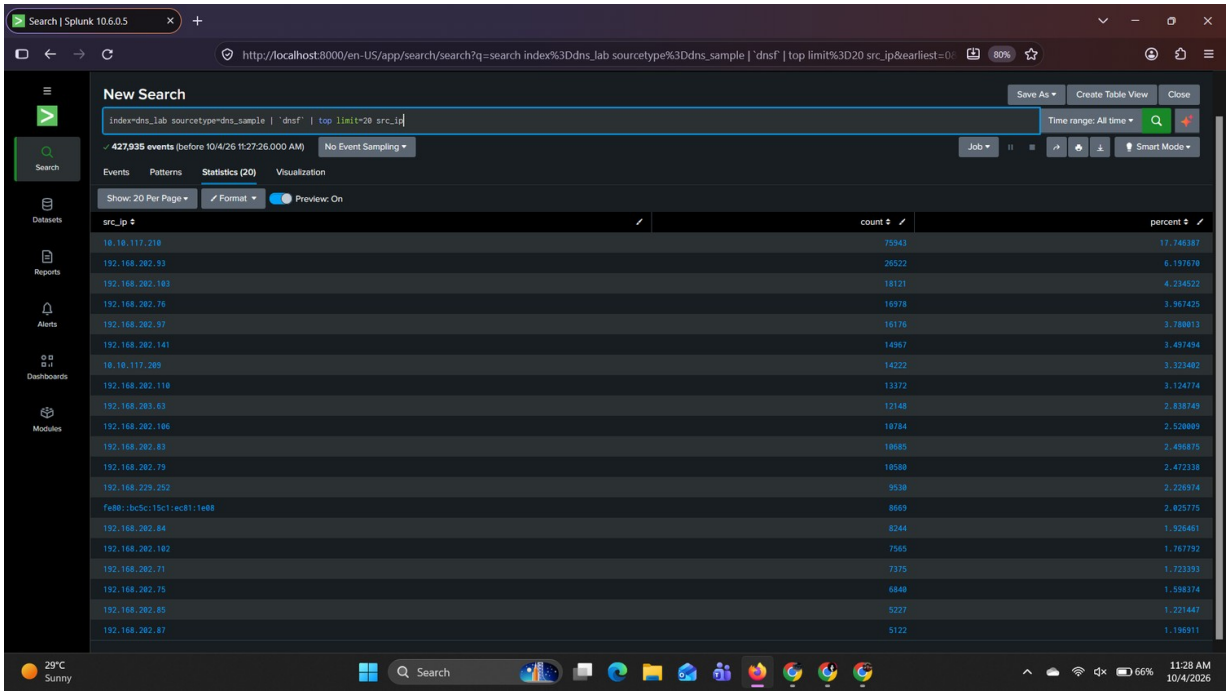


Figure 6: Top 20 source clients

The most active clients are `10.10.117.210` (75,943 queries, 17.7%), `192.168.202.93` (26,522), `192.168.202.103` (18,121), `192.168.202.76` (16,978) and `192.168.202.97` (16,176). The highest-volume client was investigated further:

SPL 7: Profile of 10.10.117.210

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=10.10.117.210 | stats count
dc(fqdn) as unique_domains min(_time) as first max(_time) as last | convert ctime(first)
ctime(last)
```

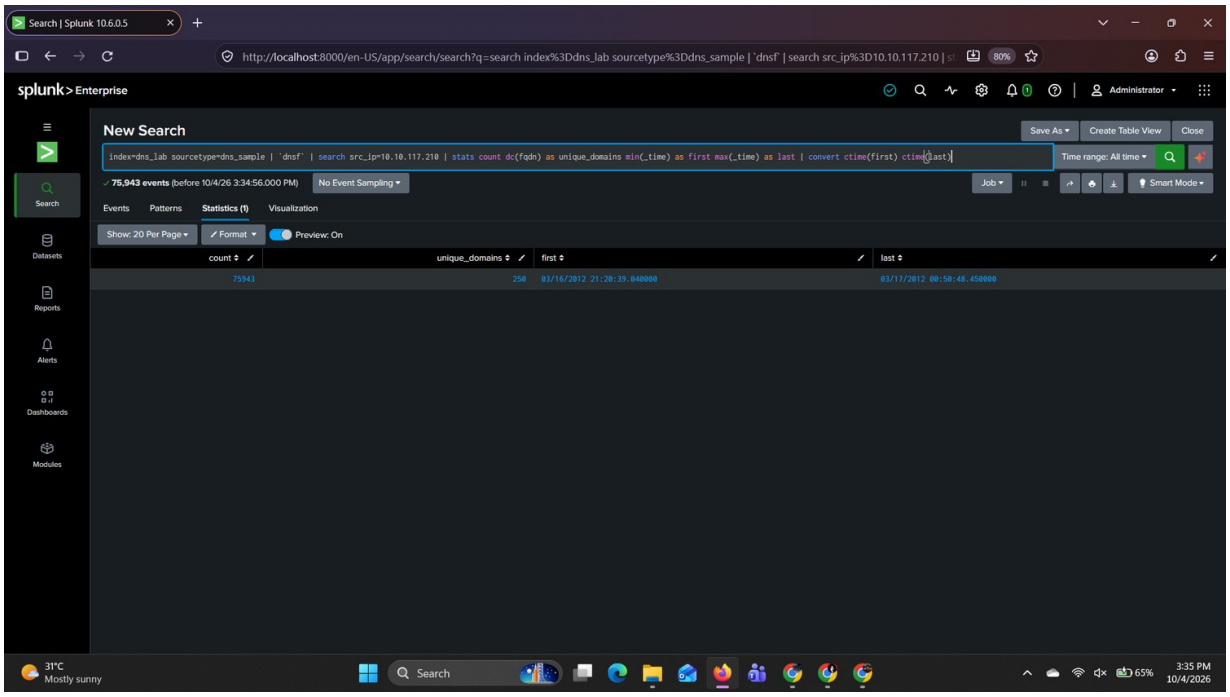


Figure 7: 10.10.117.210: event count, unique domains and active period (IST)

SPL 8: Domains, query types and responses for 10.10.117.210

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=10.10.117.210 | stats count by fqdn query_type response_code | sort - count | head 15
```

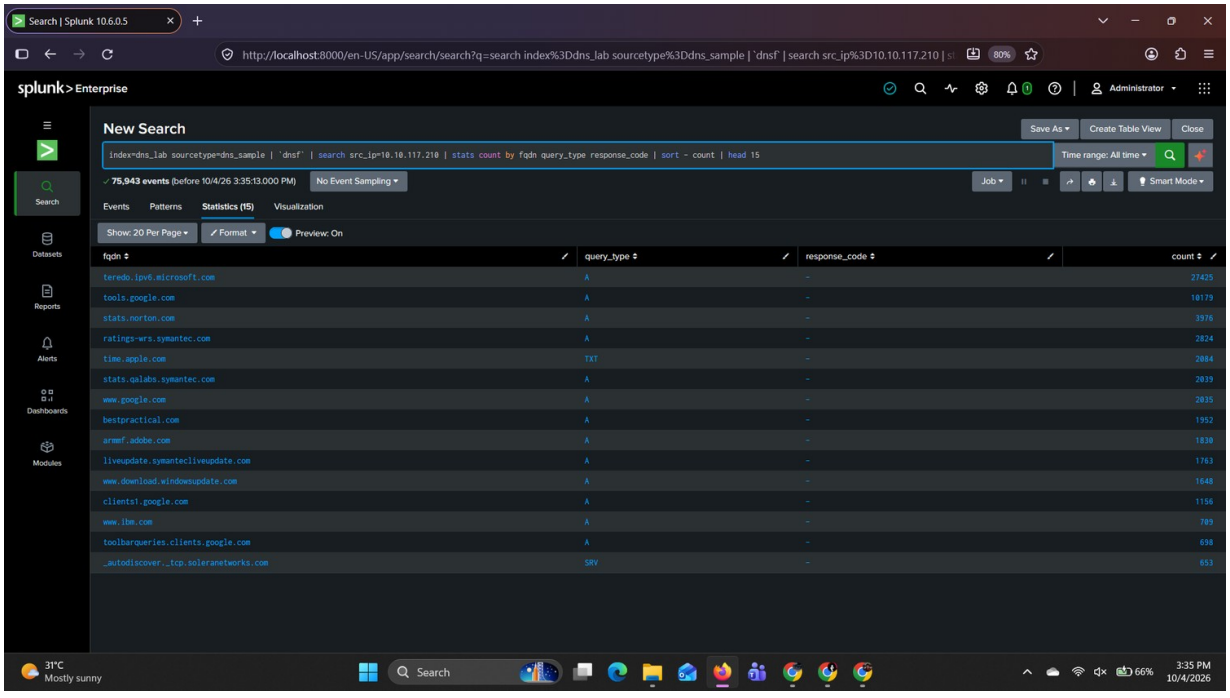


Figure 8: 10.10.117.210: most requested domains, query types and response codes

Observation	Value
Total queries / unique domains	75,943 / only 250
Active period (UTC)	16 Mar 15:50:39 to 19:20:48 (Splunk shows 21:20:39 IST to 00:50:48 IST)
Top domains	`teredo.ipv6.microsoft.com` 27,425; `tools.google.com` 10,179; `stats.norton.com` 3,976; `ratings-wrs.symantec.com` 2,824; `time.apple.com` (TXT) 2,084

Observation	Value
Response codes	All shown as `-` (no reply recorded)

Assessment: A very high volume against a small set of domains (Teredo, Google, Norton/Symantec, update services), with no NXDOMAIN failures and no scanning-style diversity, points to a gateway, proxy or a host running automated software. This client also produces the 16 March spike (Task 06). The behaviour is assessed as **expected, pending confirmation**. To confirm, I would check DHCP and asset records and the number of users behind this address.

Task 04: Query-type distribution

SPL 9

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by query_type | sort - count
```

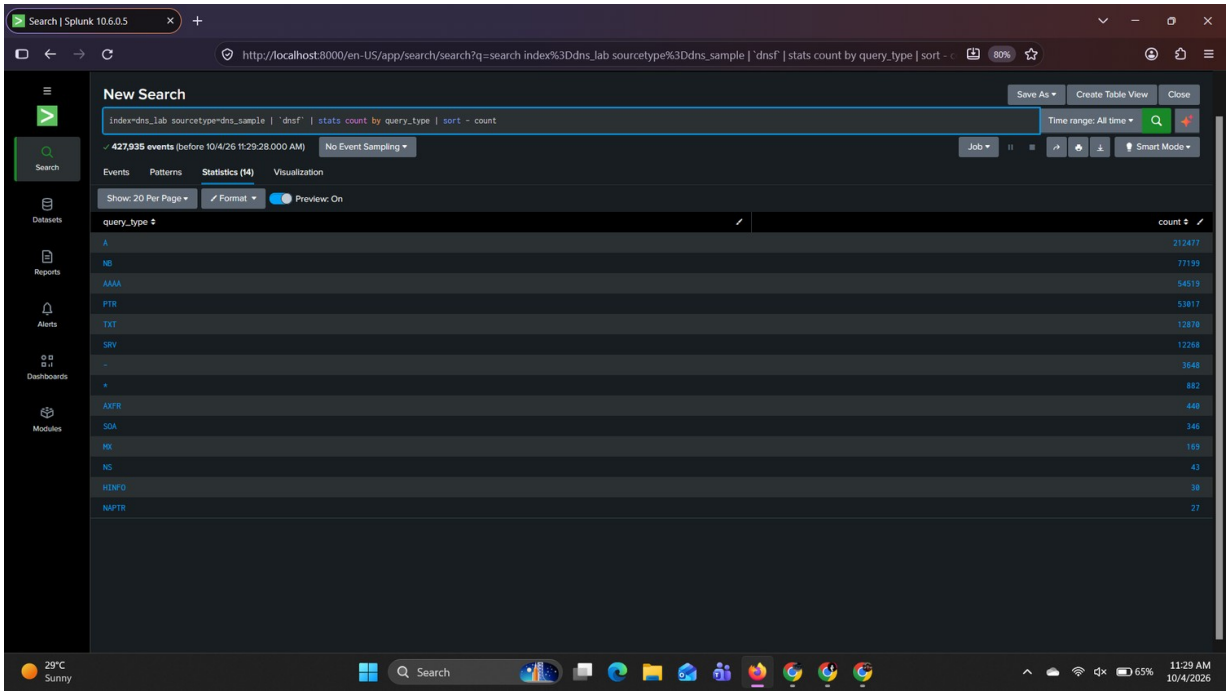


Figure 9: Query-type distribution

Dominant types: A (212,477), NB (77,199), AAAA (54,519), PTR (53,017), TXT (12,870) and SRV (12,268). NB is NetBIOS name lookup; PTR is reverse lookup.

Uncommon types that deserve review: AXFR (440), which is a request for a full zone transfer and is not normal workstation behaviour; the wildcard `*` type (882); SOA (346), MX (169), NS (43), HINFO (30) and NAPTR (27). AXFR is developed in Section 5.6.

Task 05: Response-code analysis

SPL 10

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by response_code | sort - count
```

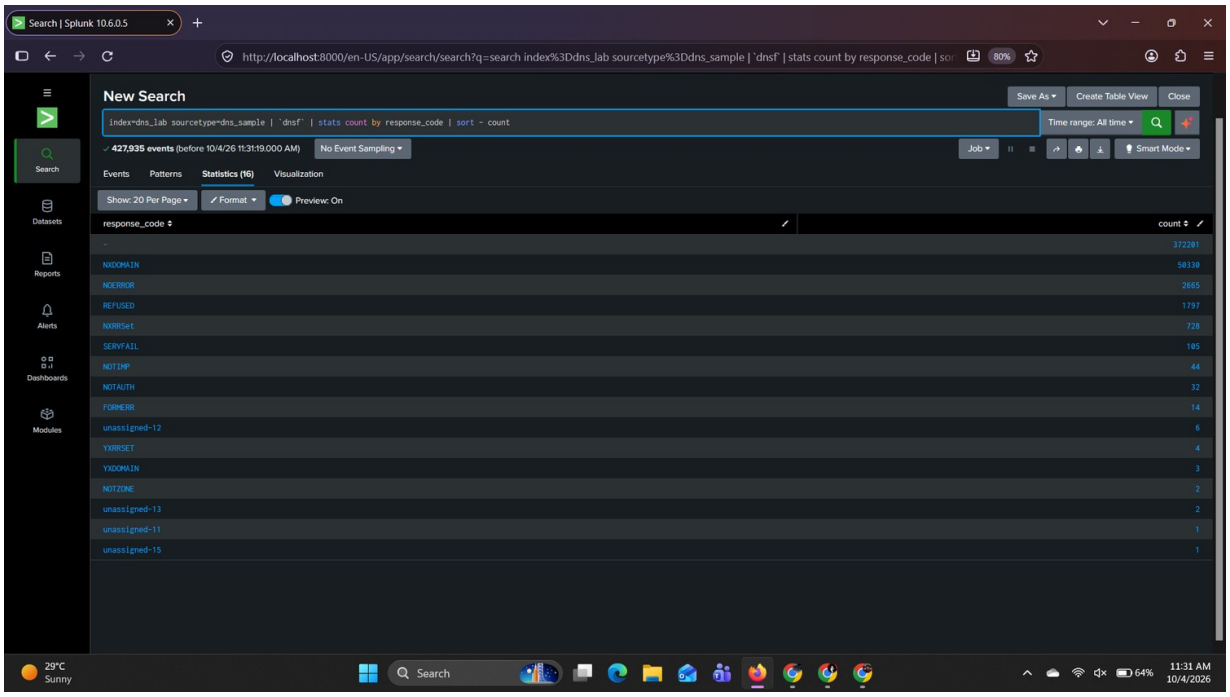


Figure 10: Response-code distribution

Response code	Count	Meaning
`-`	372,201	Query logged, no reply recorded
NXDOMAIN	50,330	Name does not exist
NOERROR	2,665	Successful answer
REFUSED	1,797	Server refused the query
NXRRSet	728	Name exists, requested record type does not
SERVFAIL / NOTIMP / NOTAUTH / FORMERR	105 / 44 / 32 / 14	Server failure, not implemented, not authoritative, malformed query
Rare codes	1 to 6 each	`unassigned-*`, YXRRSET, YXDOMAIN, NOTZONE

Concentration of failures. NXDOMAIN is concentrated in particular clients and is analysed in Section 5.3. REFUSED is also concentrated: 1,666 of the 1,797 REFUSED responses come from one client (192.168.202.88) querying one server (192.168.206.44), and about 71% of REFUSED responses involve `_dns-sd` (mDNS service-discovery) names. This looks like a misconfigured service-discovery client and not an attack.

Task 06: DNS activity over time

SPL 11

```
index=dns_lab sourcetype=dns_sample | timechart span=1h count
```

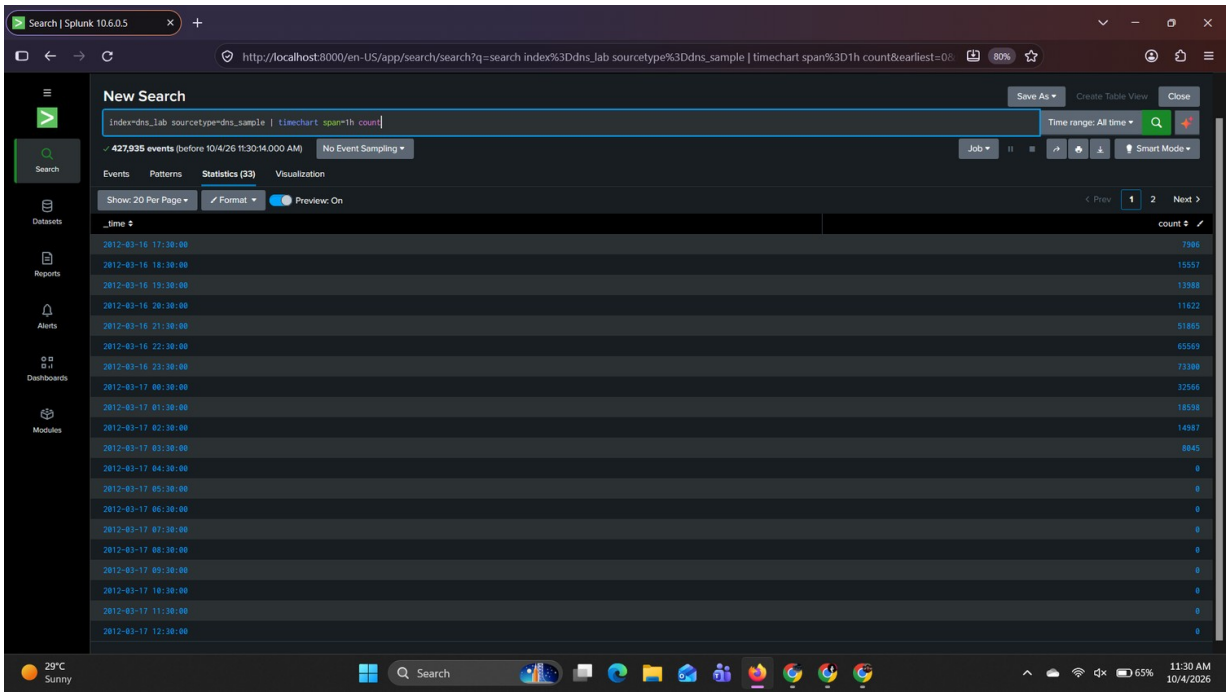


Figure 11: Hourly event counts (Splunk table, IST). Subtract 5 h 30 min for UTC

Figure 12 is a supplementary chart drawn from the same log in UTC, so the whole timeline, including the overnight gap, is visible in one view.

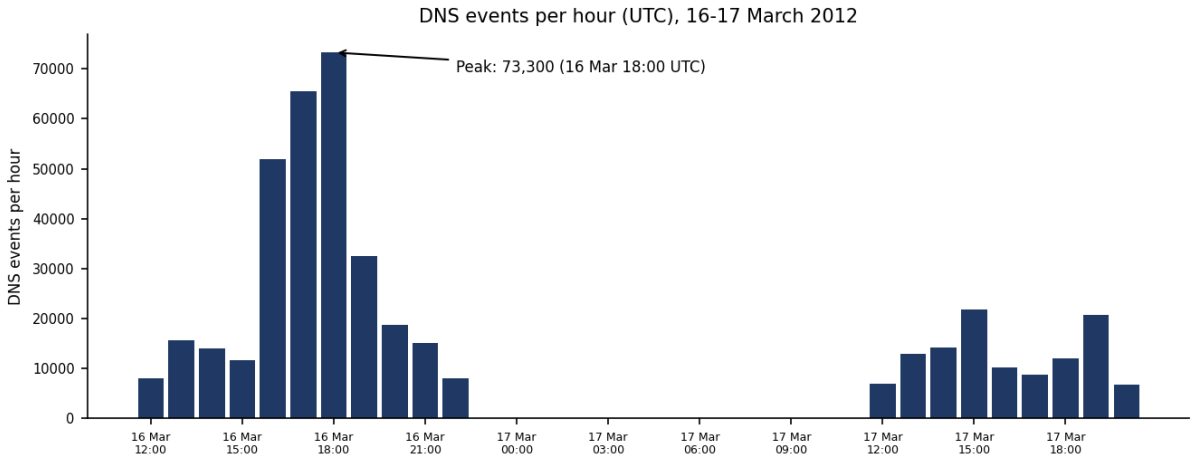


Figure 12: Supplementary chart: DNS events per hour in UTC (generated from dns.log)

Hour (UTC)	Events	Splunk shows (IST)
16 Mar 12:00	7,906	17:30
16 Mar 16:00	51,865	21:30
16 Mar 17:00	65,569	22:30
16 Mar 18:00 (peak)	73,300	23:30
16 Mar 19:00	32,566	17 Mar 00:30
16 Mar 20:00	18,598	01:30

Observations: Activity peaks on **16 March between 16:00 and 19:00 UTC**, reaching 73,300 events at 18:00 against a normal 8,000 to 20,000 per hour. This spike is driven by `10.10.117.210` (Task 03). There is also **no**

data from about 23:00 UTC on 16 March until about 12:00 UTC on 17 March, which suggests capture stopped or the network was quiet. A smaller second period of activity follows on 17 March.

Additional evidence I would inspect: the clients behind the spike (Section 5.4), query diversity and response codes, and DHCP, proxy and firewall logs for the same window.

Task 07: Top client-to-domain relationships

SPL 12

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by src_ip fqdn | sort - count | head 30
```

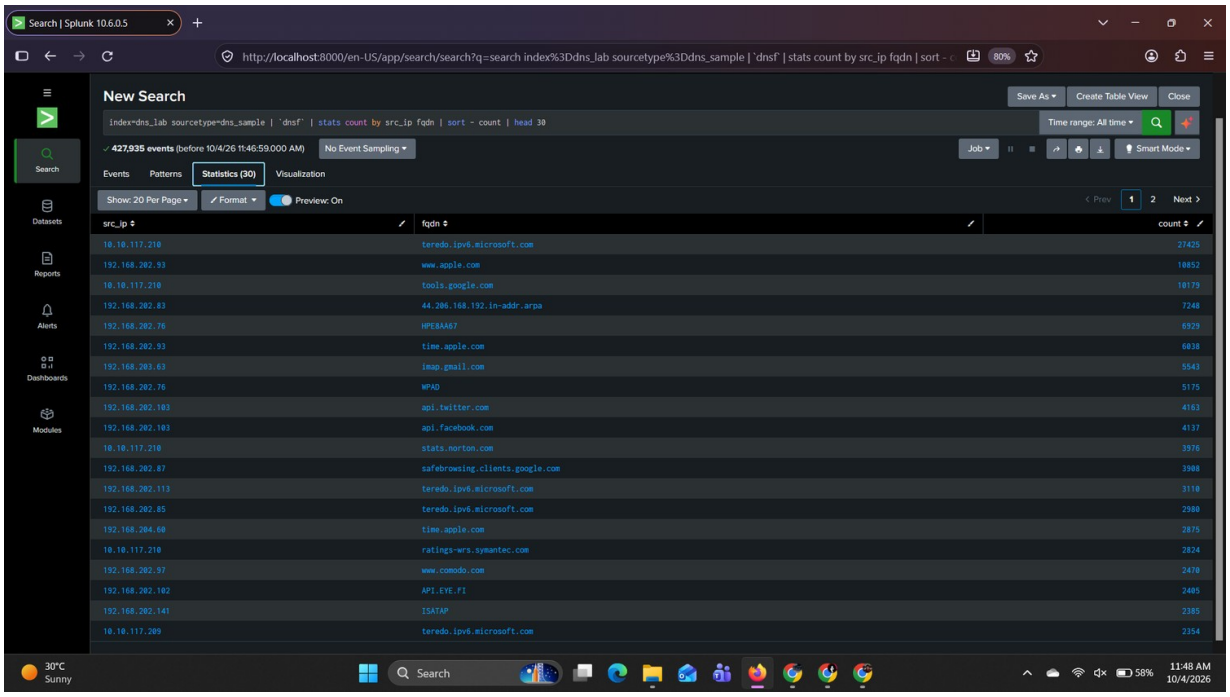


Figure 13: Top 30 client/domain pairs

Client	Domain	Queries	Review note
10.10.117.210	`teredo.ipv6.microsoft.com`	27,425	Routine tunnelling; see Task 03
192.168.202.93	`www.apple.com`	10,852	Routine
192.168.202.83	`44.206.168.192.in-addr.arpa`	7,248	Review: one reverse lookup repeated thousands of times (retry loop)
192.168.202.103	`api.twitter.com`, `api.facebook.com`	4,163 / 4,137	Review: repeated API lookups, many failing (Task 09)
192.168.202.76	`HPE8AA67`, `WPAD`	6,929 / 5,175	Local NetBIOS names

Three relationships deserve deeper review: (1) `192.168.202.83` to the single reverse-lookup name, (2) `192.168.202.103` to Twitter/Facebook API names that repeatedly fail, and (3) `192.168.202.79` to the `hec.net` zones (Section 5.6).

Task 08: Rarely queried domains

SPL 13

```
index=dns_lab sourcetype=dns_sample | `dnsf` | rare limit=30 fqdn
```

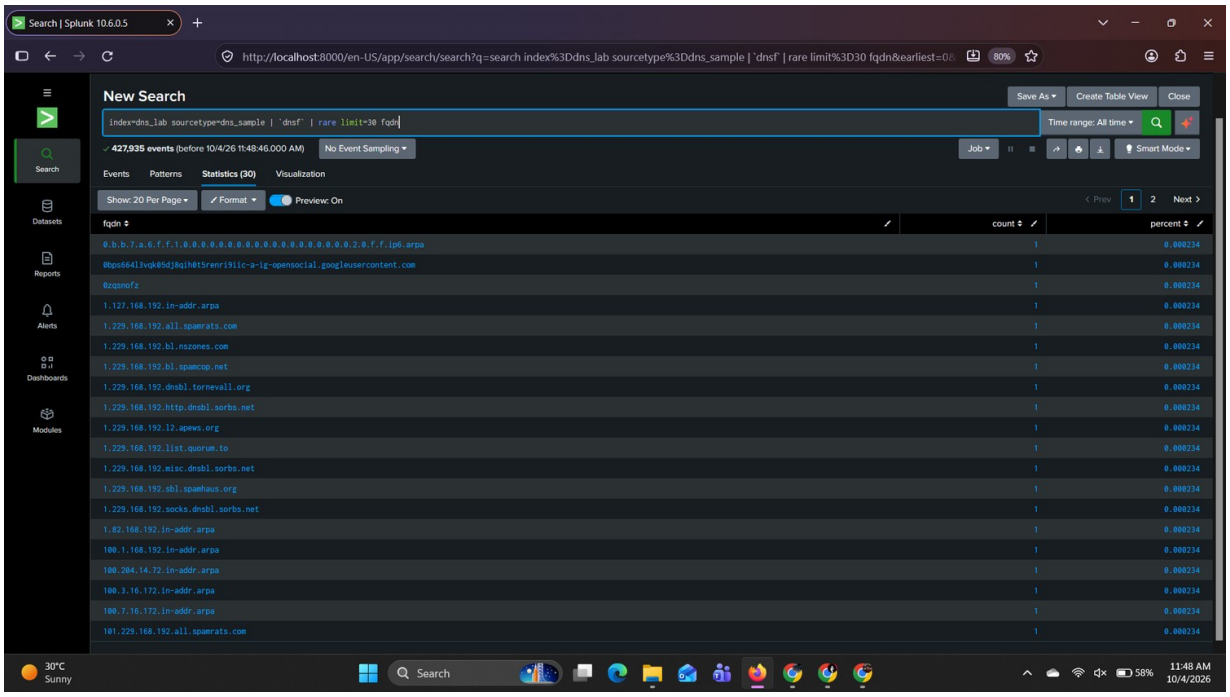


Figure 14: Rare domains (count = 1)

Rare domain	Evidence worth reviewing
`1.229.168.192.bl.spamcop.net` (and `.all.spamrats.com`, `.list.quorum.to`, `.dnsbl.sorbs.net` ...)	Internal address 192.168.229.1 checked against many public spam blocklists; typical of scanning tools, not normal browsing (Section 5.9)
`metasploitable`, `acme-texmcxhenx`	Seen once from 192.168.202.88, NXDOMAIN; look like probe or test names
`0zqsnofz`	Random single-label name, once, from 192.168.202.140, NXDOMAIN
`100.204.14.72.in-addr.arpa`	Single reverse lookup of an external address from 192.168.202.115; could be part of an address sweep
`0.b.b.7.a.6.f.f.1.0...ip6.arpa`	IPv6 reverse lookup; usually benign but rare

Rarity is a signal that prompts a closer look. It is **not** proof of maliciousness: many one-off names are typos, one-time software lookups or random browser probes.

Task 09: NXDOMAIN investigation

SPL 14

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search response_code=NXDOMAIN | stats count by src_ip fqdn | sort - count | head 30
```

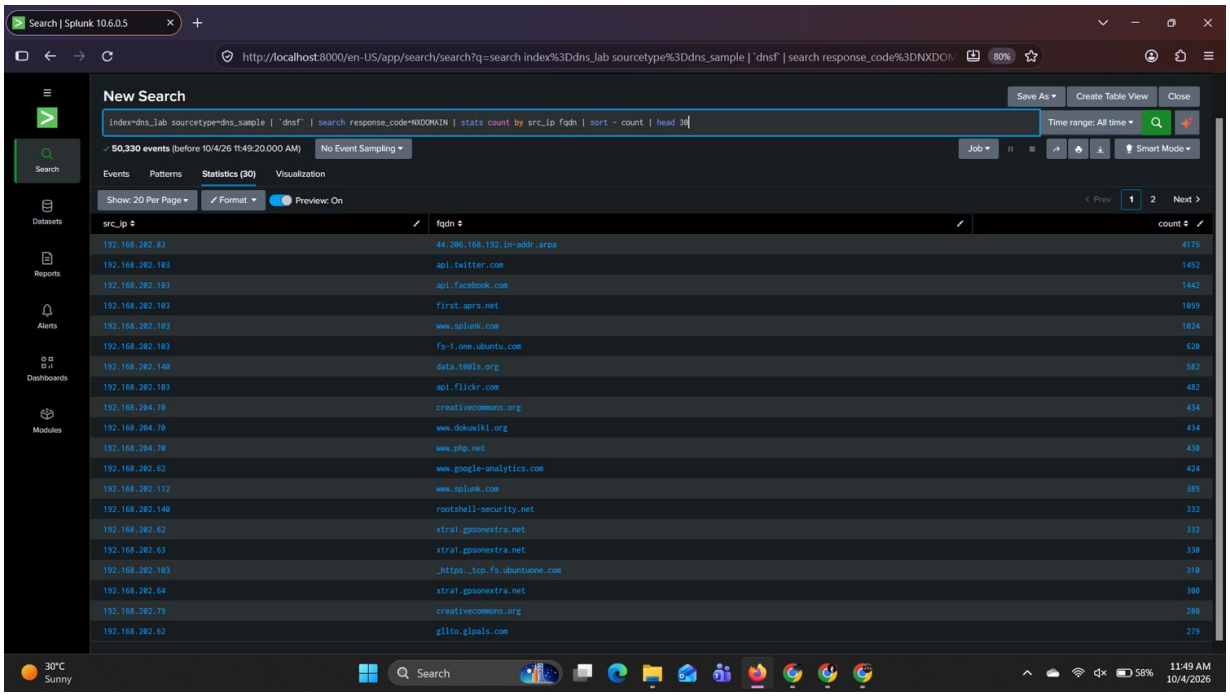


Figure 15: Top client/domain pairs returning NXDOMAIN

Client	Domain	NXDOMAIN
192.168.202.83	`44.206.168.192.in-addr.arpa`	4,175
192.168.202.103	`api.twitter.com` / `api.facebook.com` / `first.aprs.net` / `www.splunk.com`	1,452 / 1,442 / 1,059 / 1,024
192.168.202.140	`data.t00ls.org` / `rootshell-security.net`	582 / 332
192.168.204.70	`creativecommons.org` / `www.dokuwiki.org` / `www.php.net`	434 / 434 / 430

Repeated failures for the same name (`.83`, `.103`) suggest software retrying. Hosts that fail on thousands of **different** names are a different pattern and are analysed in Section 5.3.

Task 10: Suspicious keyword hunt

SPL 15

```
index=dns_lab sourcetype=dns_sample | regex _raw="(?i)(login|verify|secure|update|account|
password|wallet)" | `dnsf` | stats count by fqdn src_ip | sort - count
```

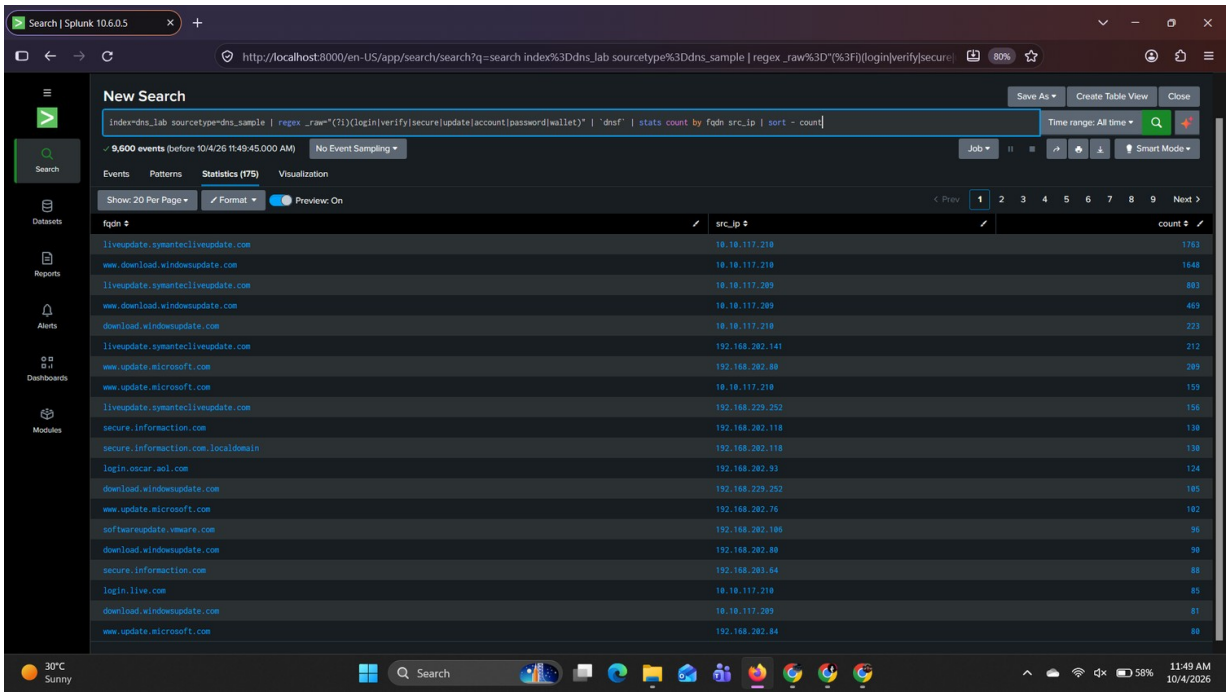


Figure 16: Keyword hunt results (9,600 events matched)

The regular expression matched **9,600 events**. Almost all are legitimate update and login services: `liveupdate.symanteclive update.com` (1,763 from 10.10.117.210), `www.download.windowsupdate.com` (1,648), `www.update.microsoft.com`, `login.live.com`, `login.oscar.aol.com` and `softwareupdate.vmware.com`.

One name needs validation: **`secure.informaction.com`** (874 queries from 29 clients, plus a `.localdomain` variant of 130 queries from 192.168.202.118, which suggests the host's resolver appended its local search domain). Because many clients query it, it is probably shared software or a service, but it should be checked against threat intelligence before closing. No obvious phishing-style names were found. **No escalation is required on the keyword results alone.**

5. Investigation Findings

5.1 High-volume clients versus unique domains (Phase 3A)

SPL 16

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count as dns_queries dc(fqdn) as
unique_domains by src_ip | sort - dns_queries | head 20
```

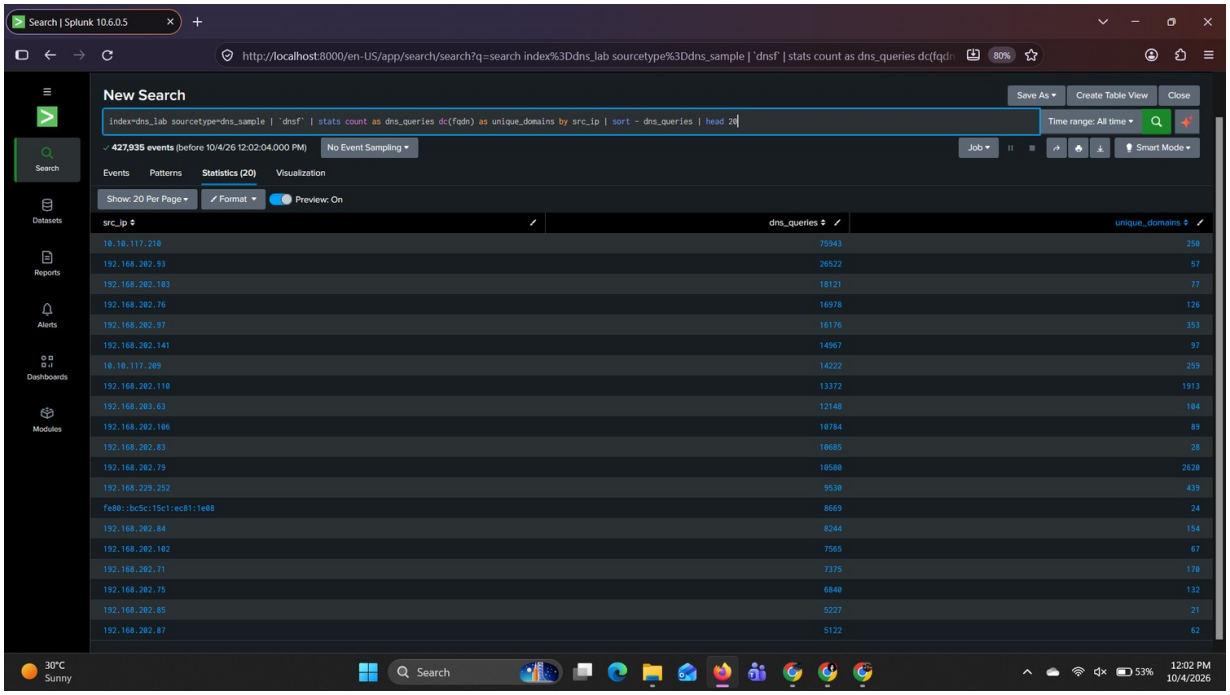


Figure 17: Query volume compared with unique domains per client

Client	Queries	Unique domains	Reading
10.10.117.210	75,943	250	High volume, low diversity: repeated names
192.168.202.93	26,522	57	Repeated names
192.168.202.103	18,121	77	Repeated names
192.168.202.110	13,372	1,913	High diversity: review
192.168.202.79	10,580	2,620	Highest diversity of any client: strongest lead
192.168.229.252	9,530	439	Moderate

A client that contacts thousands of different domains from a modest number of queries behaves differently from a client that repeats a few names. `192.168.202.79` queried 2,620 unique domains from 10,580 events (about one new name for every four queries).

5.2 Domains contacted by many clients (Phase 3B)

SPL 17

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats dc(src_ip) as unique_clients count as
queries by fqdn | sort - unique_clients | head 30
```

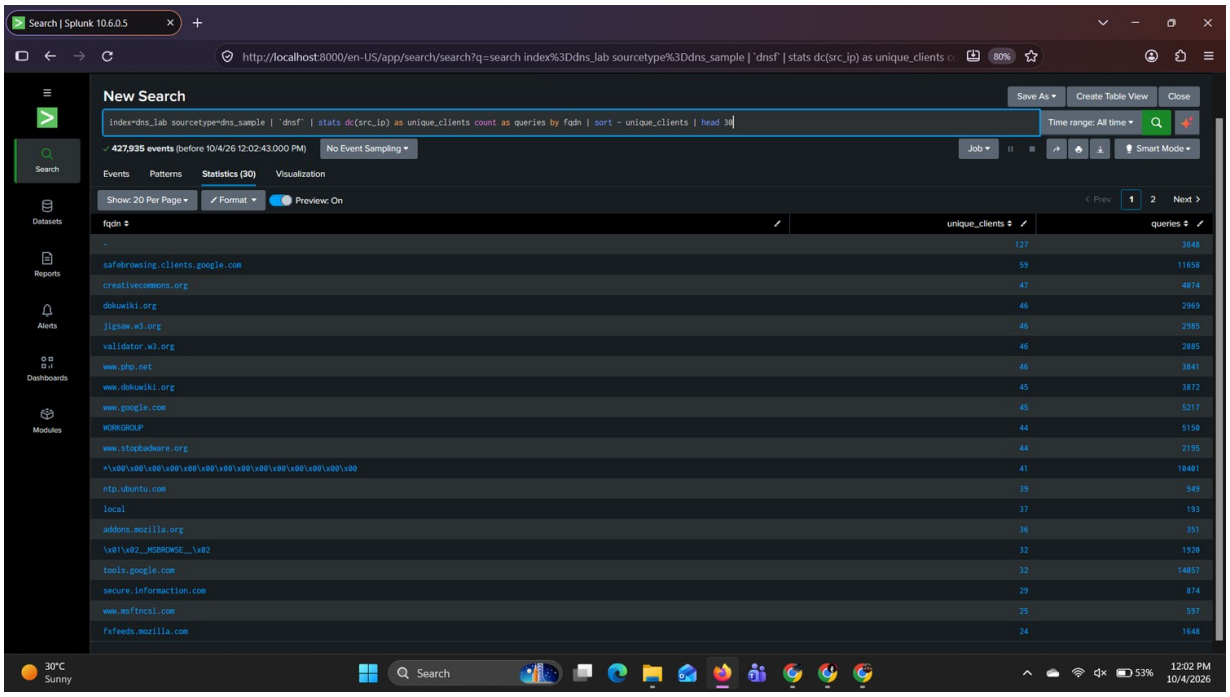


Figure 18: Domains ranked by number of distinct clients

The domains with the most distinct clients are common infrastructure: `safebrowsing.clients.google.com` (59 clients), `creativecommons.org` (47), `dokuwiki.org`, `jigsaw.w3.org`, `validator.w3.org` and `www.php.net` (46 each), `www.google.com` (45) and `ntp.ubuntu.com` (39). The `-` entry (127 clients) is not a domain: it marks records with no query name. `secure.information.com` (29 clients, 874 queries) is shared software or a service and is listed for validation.

5.3 Repeated NXDOMAIN by client (Phase 3C)

SPL 18

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search response_code=NXDOMAIN | stats count as nxdomain_count dc(fqdn) as unique_failed_domains by src_ip | sort - nxdomain_count
```

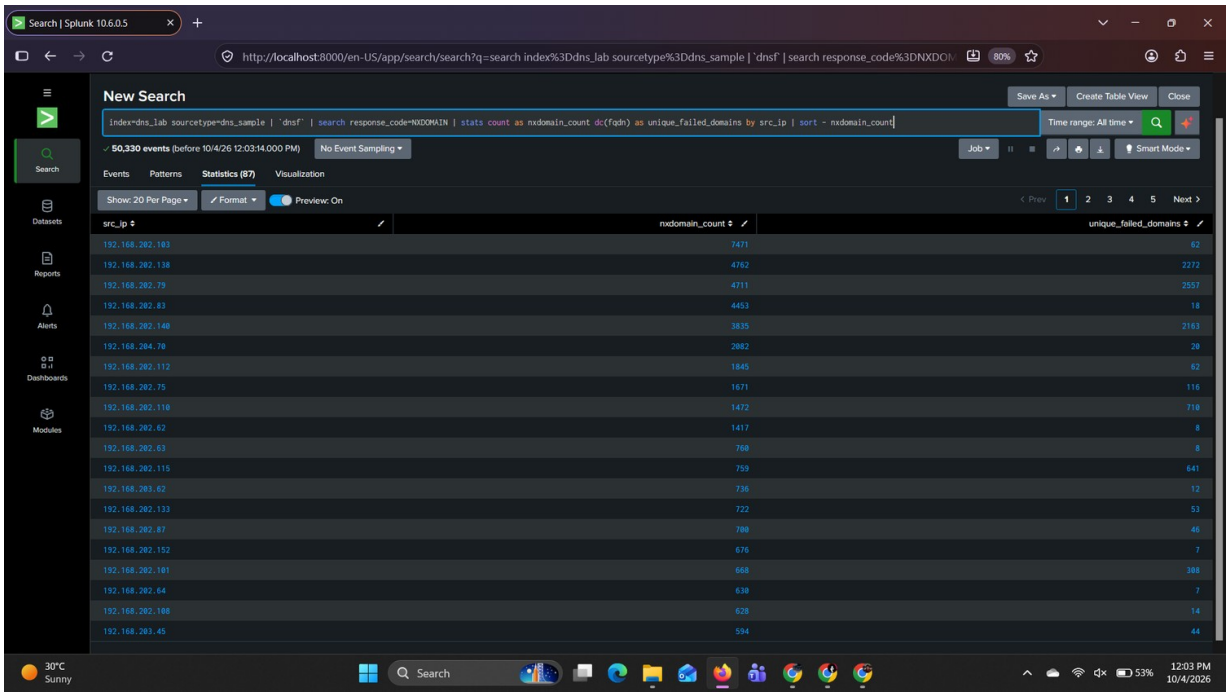


Figure 19: NXDOMAIN count compared with unique failed domains per client

Client	NXDOMAIN	Unique failed domains	Reading
192.168.202.103	7,471	62	Retrying a few names: benign pattern
192.168.202.83	4,453	18	Retrying one reverse lookup: benign pattern
192.168.202.138	4,762	2,272	Many different failed names: scanning pattern
192.168.202.79	4,711	2,557	Many different failed names: scanning pattern
192.168.202.140	3,835	2,163	Many different failed names: scanning pattern
192.168.202.110 / .115	1,472 / 759	710 / 641	Elevated diversity: secondary leads

The key distinction is the ratio of unique failed names to total failures. For `.79`, `.138` and `.140`, about half of all failures are for a *different* name (48% to 56%). For `.103` and `.83` it is below 1%. A high NXDOMAIN rate can be benign (software retries, typos), so it is treated as a lead and correlated with other evidence in Section 5.6 onward.

5.4 Query bursts (Phase 3D)

SPL 19

```
index=dns_lab sourcetype=dns_sample | `dnsf` | bin _time span=5m | stats count as queries by _time src_ip | sort - queries | head 50
```

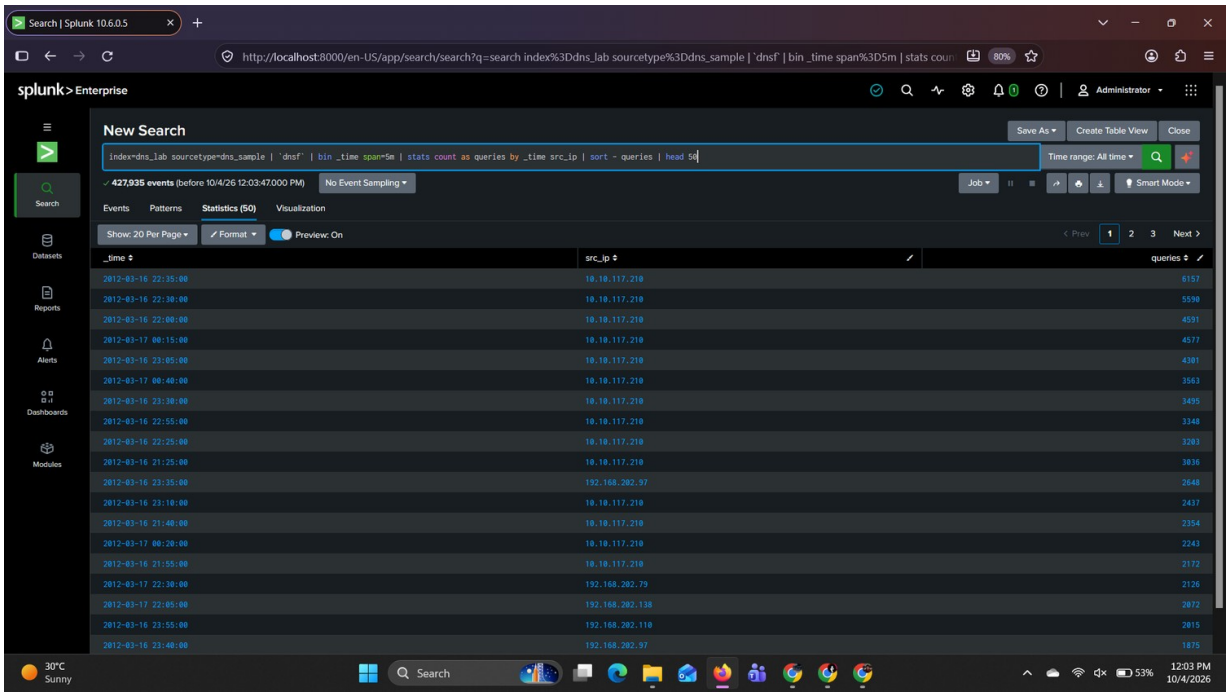


Figure 20: Largest 5-minute query bursts by client (Splunk shows IST)

The largest burst is `10.10.117.210` with **6,157 queries in five minutes** (17:05 UTC on 16 March; shown as 22:35 IST). This client sustains bursts from about 16:00 to 19:00 UTC, which explains the spike in Task 06. On 17 March the largest bursts come from `192.168.202.79` (2,126 queries at 17:00 UTC) and `192.168.202.138` (2,072 at 16:35 UTC), the same hosts that show high unique-failure counts. Burst volume is therefore consistent with automated activity on those hosts.

5.5 Domain pivot: rootshell-security.net (Phase 3E)

SPL 20

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn="rootshell-security.net" | stats count values(src_ip) as clients values(query_type) as query_types by fqdn
```

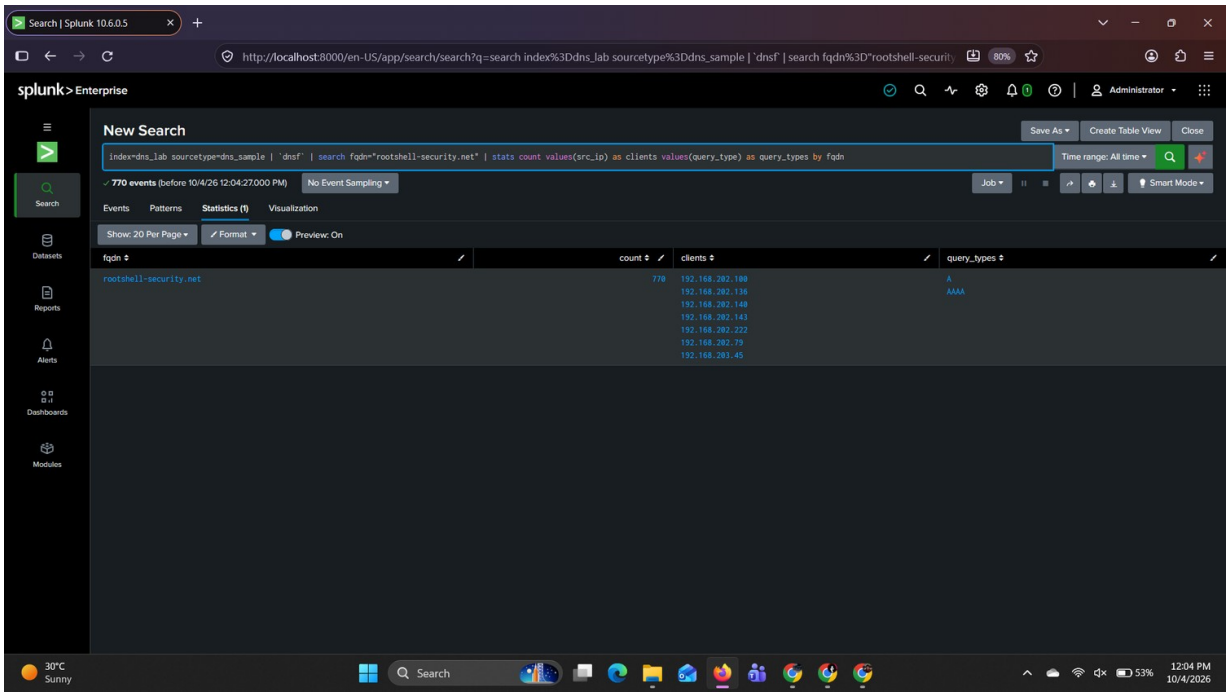


Figure 21: Clients requesting rootshell-security.net

Client	Queries
192.168.202.140	332
192.168.202.79	228
192.168.202.143	106
192.168.202.136	72
192.168.202.100 / .222 / 192.168.203.45	16 / 8 / 8

The domain was requested **770 times by seven clients**, using A and AAAA records. **All 770 responses were NXDOMAIN**, so the name never resolved. All requests fall within one hour on 17 March (19:24 to 20:23 UTC). The group includes `192.168.202.79` and `192.168.202.140`, two hosts flagged for scanning behaviour. Several clients asking for the same unresolved name in the same hour could indicate a shared tool, script or web page, but this alone is not evidence of compromise. The domain is validated in Section 6.

5.6 Notable observation 1: Zone-transfer (AXFR) requests

Finding

Host **192.168.202.79** sent **437 of the 440 AXFR requests**. A zone transfer asks a DNS server for its complete list of records and is not normal workstation behaviour. **415 requests received NOERROR responses containing internal hostnames and IP addresses.**

SPL 21: AXFR requests by client

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search query_type=AXFR | stats count dc(fqdn) as zones dc(dest_ip) as dns_servers min(_time) as first max(_time) as last by src_ip | convert ctime(first) ctime(last)
```

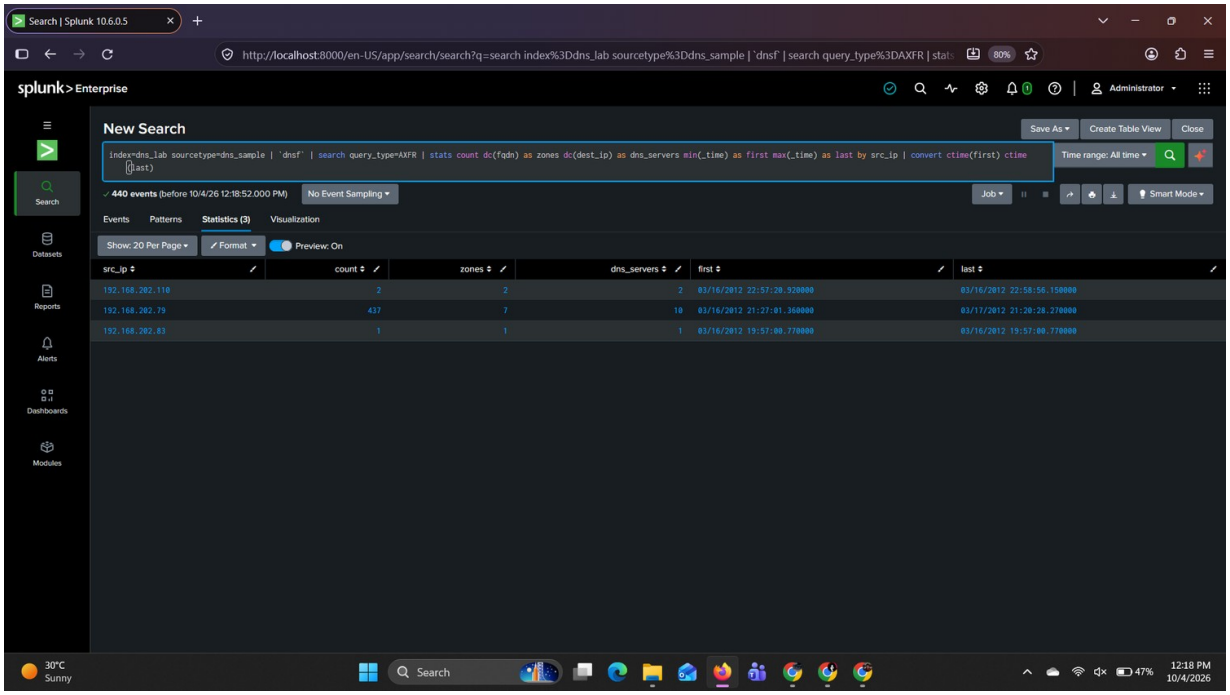


Figure 22: AXFR requests by source host (Splunk shows IST)

Source	AXFR requests	Zones	DNS servers	First / last (UTC)
192.168.202.79	437	7	10	16 Mar 15:57:01 to 17 Mar 15:50:28
192.168.202.110	2	2	2	16 Mar 17:27 to 17:28
192.168.202.83	1	1	1	16 Mar 14:27

SPL 22: Zones and servers targeted

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR | stats count by fqdn dest_ip | sort - count
```

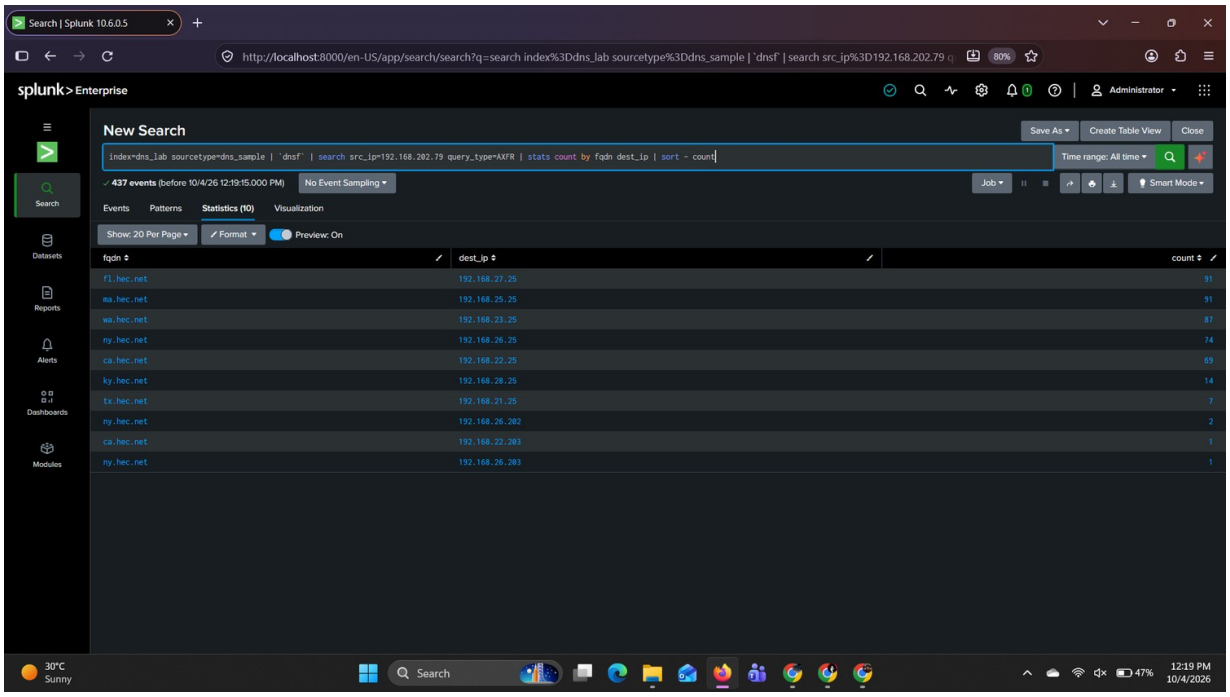


Figure 23: Zones and DNS servers targeted by 192.168.202.79

The host requested seven zones (`fl`, `ma`, `wa`, `ny`, `ca`, `ky`, `tx.hec.net`). Each zone was requested from its own site DNS server `192.168.2N.25` (N = 21 to 28), for example `fl.hec.net` from 192.168.27.25 (91 requests), `ma.hec.net` from 192.168.25.25 (91), `wa.hec.net` from 192.168.23.25 (87), `ny.hec.net` from 192.168.26.25 (74) and `ca.hec.net` from 192.168.22.25 (69). Requesting every site's zone from that site's server is systematic enumeration, not a one-off administrative query. All 437 requests used **TCP port 53**, the transport zone transfers use.

SPL 23: AXFR responses with answer data

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR
response_code=NOERROR | rex field=_raw "\t(?<answers>[^\t]+)\t(?<TTLs>[^\t]+)\t(?
<rejected>[^\t]+)$" | table _time fqdn dest_ip answers | head 10
```

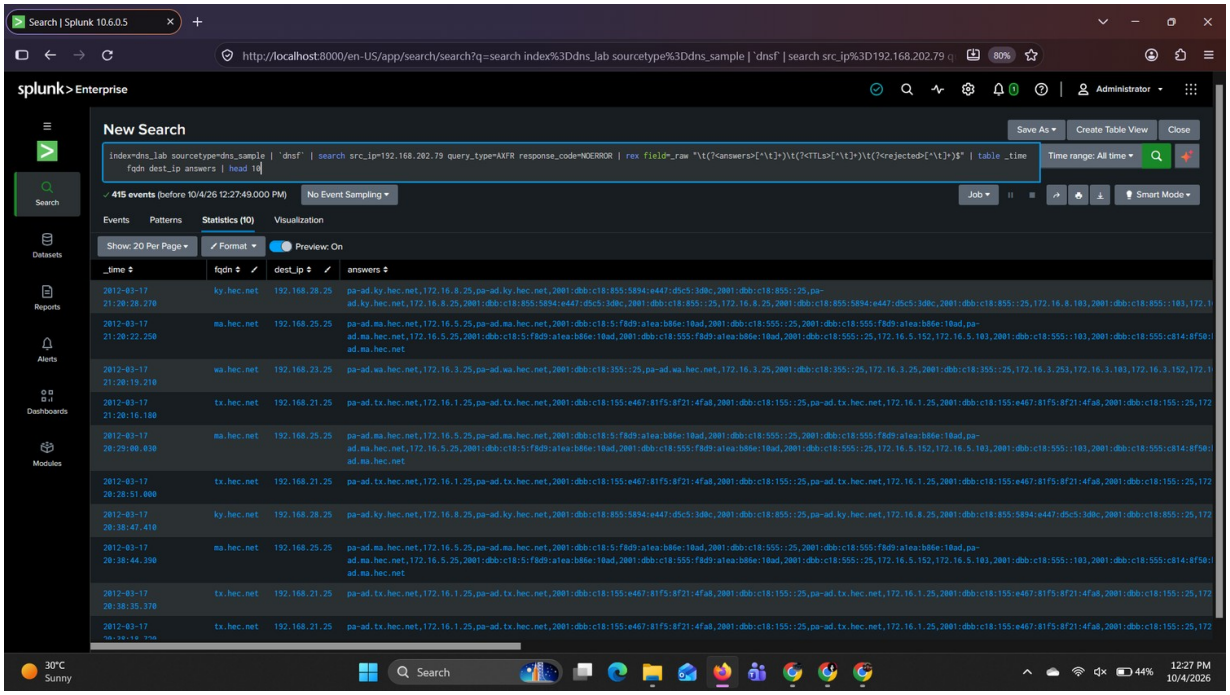


Figure 24: AXFR responses (NOERROR) containing internal hostnames and addresses (Splunk shows IST)

Of the 437 requests, **415 returned NOERROR**, 16 had no recorded reply, 4 returned NOTAUTH and 2 returned REFUSED. The `answers` field of the NOERROR responses holds real records: for example `pa-ad.ky.hec.net` with `172.16.8.25` and `172.16.8.103` plus IPv6 addresses, and equivalent records for the `ma`, `wa` and `tx` sites. The DNS servers therefore **appear to allow zone transfers to an ordinary workstation**, which exposes an internal host inventory.

Preserved raw evidence

SPL 24: Raw AXFR events

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR
| head 3 | table _time _raw
```

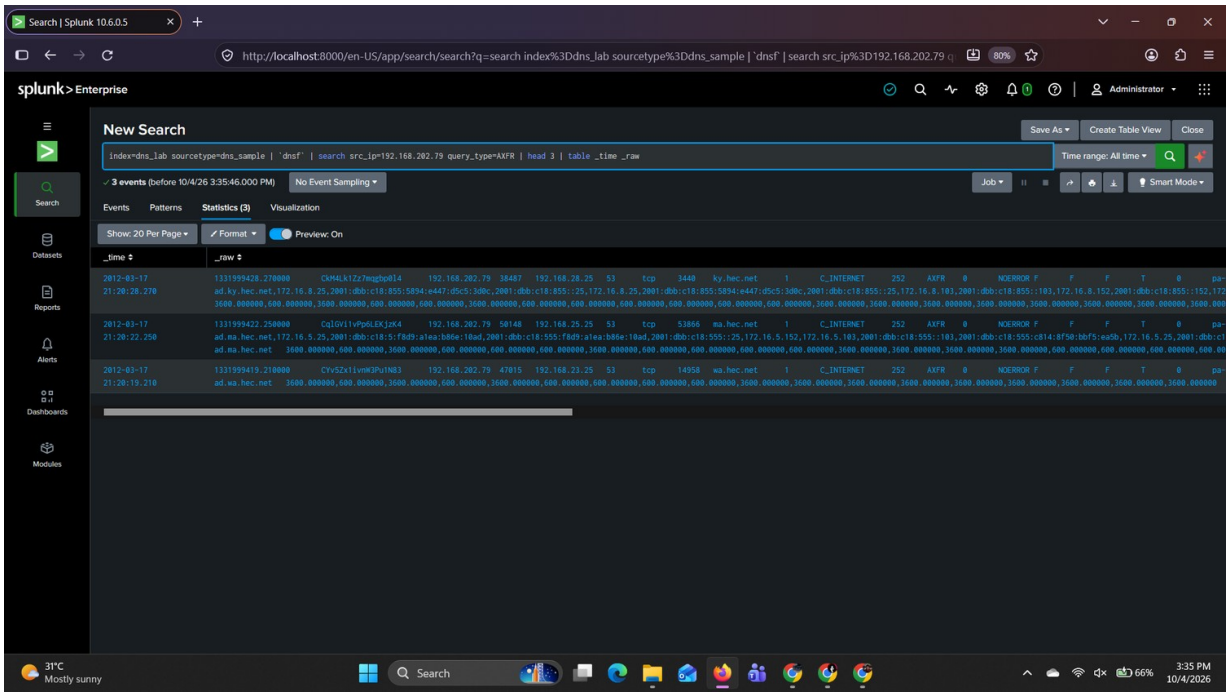


Figure 25: Raw AXFR events preserved with timestamps

Raw field (first event)	Value	Meaning
`ts`	1331999428.270000	17 Mar 2012 15:50:28.270 UTC
`uid`	CkM4Lk1Zz7mqgbp0l4	Zeek connection ID
source	192.168.202.79 : 38487	Client and client port
destination	192.168.28.25 : 53	DNS server, port 53
`proto`	tcp	Zone transfers use TCP
query	`ky.hec.net`, class IN, type 252 (AXFR)	Zone requested
rcode	0 (NOERROR)	Server answered
flags	AA=F, TC=F, RD=F, RA=T	Authoritative/truncated/recursion flags
`answers`	pa-ad.ky.hec.net, 172.16.8.25, …	Records returned by the server

Assessment (not proven): this pattern is consistent with reconnaissance. **Limitation:** DNS logs cannot prove the complete zone was transferred or how much data moved. Firewall, `conn.log` or DNS-server logs are needed. **Possible benign explanation:** an approved vulnerability scan or penetration test.

5.7 Notable observation 2: Reverse-lookup (PTR) sweep

SPL 25: PTR queries by subnet

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=PTR | rex field=fqdn "^\d+\.(?<third_octet>\d+)\.168\.192\.in-addr\.arpa" | stats count by third_octet | sort - count
```

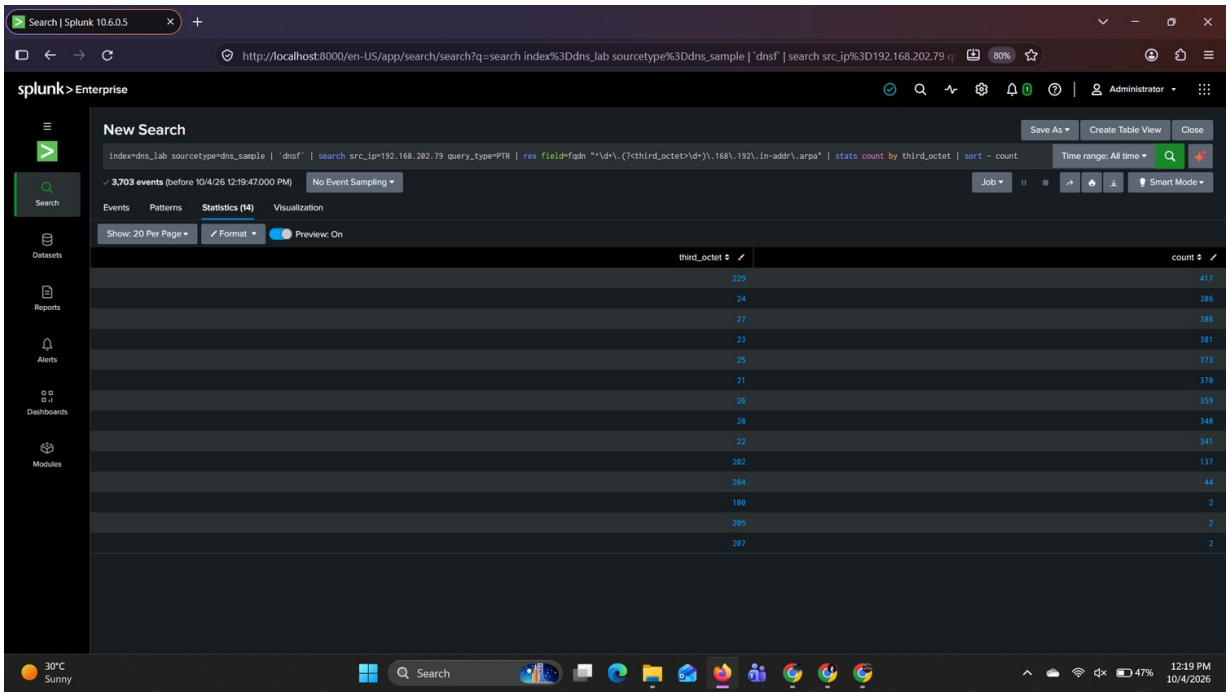


Figure 26: PTR queries from 192.168.202.79 by subnet

Subnet (192.168.x.0)	PTR queries
229	417
24 / 27 / 23	386 / 386 / 381
25 / 21 / 26	373 / 370 / 359
28 / 22	348 / 341
202 (own subnet)	137
204	44

`192.168.202.79` sent **3,703 PTR queries**, spread almost evenly across **nine subnets**: 21 to 28 (341 to 386 each) and 229 (417), against 137 for its own subnet. Walking through consecutive subnets with similar counts is characteristic of host discovery, meaning mapping which addresses exist.

SPL 26: Query types and responses for 192.168.202.79

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 | stats count by query_type response_code | sort - count
```

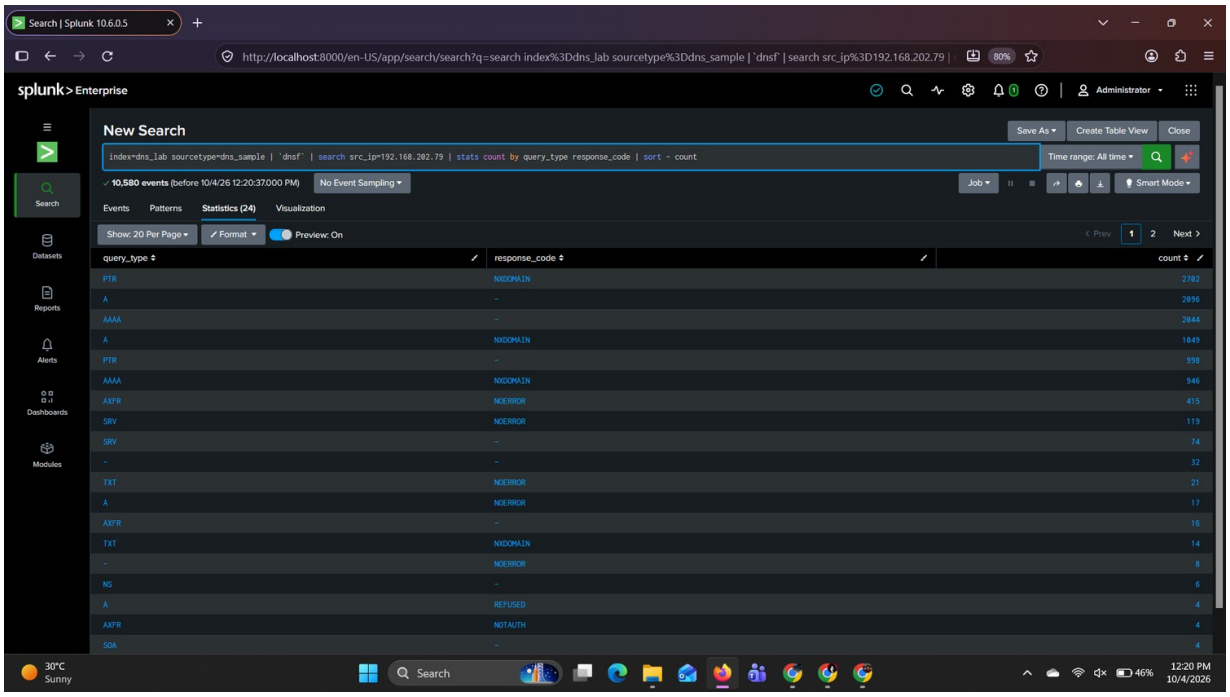


Figure 27: Query types and response codes for 192.168.202.79

Of this host's 10,580 events, **2,702 were PTR queries answered NXDOMAIN**, 1,049 were A queries answered NXDOMAIN and 946 were AAAA queries answered NXDOMAIN, alongside the 415 successful AXFR responses. Mostly failing lookups across many names, combined with successful zone transfers, show a host enumerating the network. The host was active from 16 Mar 12:30 UTC to 17 Mar 20:38 UTC.

Possible benign explanation: a network scanner or asset-discovery tool run by IT. **Evidence needed:** DHCP and asset records, the scan schedule and firewall logs.

5.8 Notable observation 3: DNS fingerprinting (version.bind)

SPL 27

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn="version.bind" | stats count
dc(dest_ip) as dns_servers by src_ip | sort - count
```

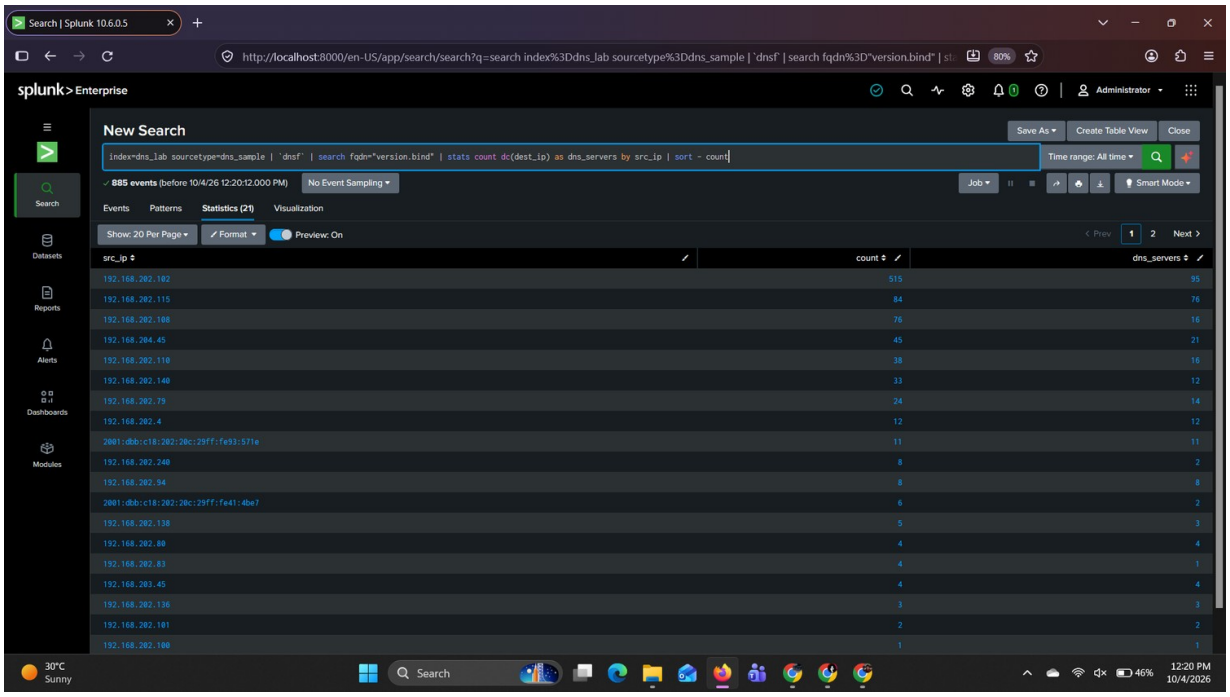


Figure 28: version.bind queries by client

Client	Queries	DNS servers asked
192.168.202.102	515	95
192.168.202.115	84	76
192.168.202.108	76	16
192.168.204.45	45	21
192.168.202.110	38	16
192.168.202.140	33	12
192.168.202.79	24	14

A `version.bind` query asks a DNS server which software and version it runs. There were **885 such queries**. `192.168.202.102` asked **95 different DNS servers** (515 queries) and `192.168.202.115` asked 76. A normal user has no reason to do this. Asking many servers for their version is consistent with fingerprinting. Several of these hosts appear elsewhere in this report (`.79`, `.108`, `.110`, `.115`, `.140`), which suggests a shared tool or an approved scan run from several machines. **`192.168.202.102` is a new host of interest.**

5.9 Notable observation 4: DNS blocklist (DNSBL) lookups

SPL 28

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn IN
("*sorbs.net","*spamhaus.org","*spamcop.net","*spamrats.com","*quorum.to","*tornevall.org","
*apews.org","*nszones.com") | stats count dc(fqdn) as lookups by src_ip | sort - count
```

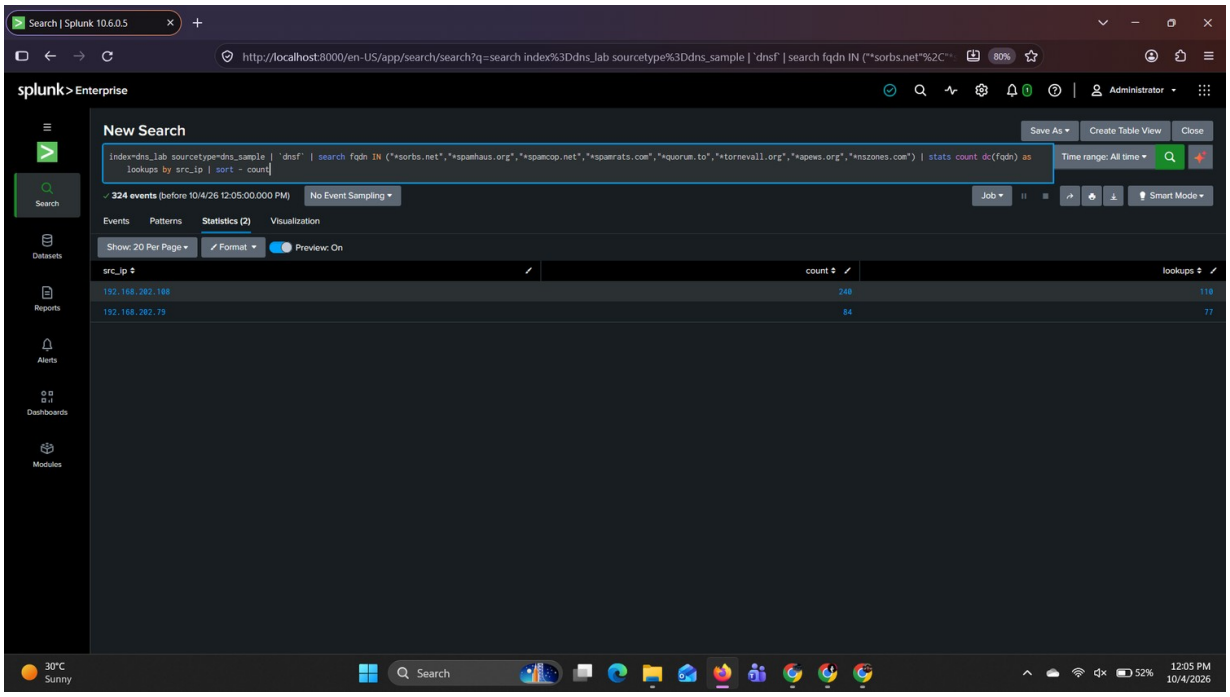


Figure 29: DNSBL lookups by source host

Client	DNSBL events	Distinct lookups
192.168.202.108	240	110
192.168.202.79	84	77

324 events came from only two hosts. The lookups follow the DNSBL format of a reversed IP address in front of a blocklist domain, for example `1.229.168.192.bl.spamcop.net` (a check of 192.168.229.1 against SpamCop). Normal users do not check addresses against many blocklists. The same host (`.79`) also sent the zone transfers and the subnet sweep, which strengthens that case. `192.168.202.108` is a second host of interest (it also sent 76 `version.bind` queries).

Possible benign explanation: mail servers and mail gateways query blocklists routinely, so `.108` could be a mail system. This is not confirmed. **Evidence needed:** asset records, and SMTP (port 25) traffic from `.108` in firewall logs.

6. Threat Intelligence Validation

Indicator: `rootshell-security.net`. **Why chosen:** requested 770 times by seven internal clients, including 192.168.202.79 and 192.168.202.140; every query returned NXDOMAIN (Section 5.5). **Date of checks:** 4 October 2026.

Source	Date	Result
VirusTotal	4 Oct 2026	4 of 91 security vendors flagged the domain: ADMINUSLabs, alphaMountain.ai and Chong Lua Dao as malicious, Fortinet as malware. Other vendors shown (for example Abusix, Acronis, AlienVault, BitDefender) reported clean. Registrar: Stichting Registrar of Last Resort Foundation; created about 2 years ago.
URLhaus	4 Oct 2026	No malware URLs listed for `rootshell-security.net`.

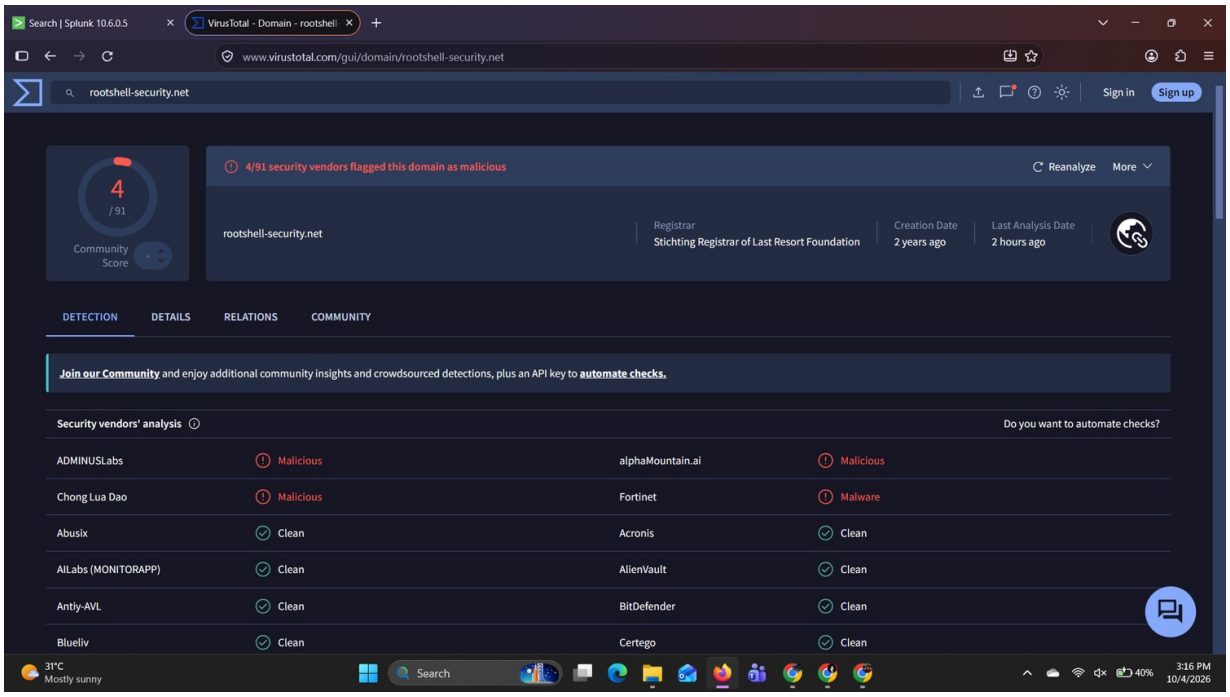


Figure 30: VirusTotal result for rootshell-security.net (checked 4 Oct 2026)

[[Insert URLhaus search-result screenshot here: search box showing rootshell-security.net with no entries]]

Figure 31: URLhaus search result for rootshell-security.net (no entries)

Interpretation: mixed and inconclusive

Four of 91 vendors flag the domain, and URLhaus has no entry. URLhaus only lists URLs used to distribute malware, so absence there does not prove a domain is safe.

Important limitation: the domain was registered about two years before these checks, but the DNS log is from **March 2012**. Today's reputation therefore describes the domain as it is now and does not show who owned or used the name in 2012. In this dataset every query for it returned NXDOMAIN, so it never resolved.

The domain is **not** labelled malicious. The case against 192.168.202.79 rests on its internal behaviour (Section 9), not on this domain.

7. Evidence Screenshots

The screenshots are placed beside the finding they support. The key evidence is indexed below.

Figure	Evidence	Supports
Figure 1	Validation: 2012 timestamps and raw records	Ingestion and validation
Figure 4	Event count and time range	Baseline
Figure 12	Hourly activity and 16 Mar spike	Task 06
Figure 8	Profile of the busiest client	Task 03
Figure 19	NXDOMAIN versus unique failed names	Scanning pattern
Figure 22 to Figure 25	AXFR requests, targets, answers and raw events	Main finding
Figure 26	Reverse-lookup sweep across nine subnets	Enumeration
Figure 28	version.bind fingerprinting	Related hosts
Figure 29	Blocklist lookups	Supporting behaviour
Figure 30	Threat-intelligence check	Phase 4

8. Answers to Investigation Questions

1. How many total DNS events are present?

427,935 events (Figure 4).

2. Earliest and latest event timestamp?

Earliest **16 Mar 2012 12:30:05 UTC**; latest **17 Mar 2012 20:59:51 UTC**. Splunk displayed 18:00:05 and 02:29:51 (18 Mar) because the user time zone is IST.

3. Which client generated the most queries?

10.10.117.210 with 75,943 queries (17.7%). It repeats only 250 domains and is assessed as expected behaviour pending asset confirmation (Task 03).

4. Which 10 domains were queried most?

`teredo.ipv6.microsoft.com` (39,273), `tools.google.com` (14,057), `www.apple.com` (13,390), `time.apple.com` (13,109), `safebrowsing.clients.google.com` (11,658), the NetBIOS wildcard name (10,401), `WPAD` (9,134), `44.206.168.192.in-addr.arpa` (7,248), `HPE8AA67` (6,929) and `ISATAP` (6,569).

5. Which client queried the most unique domains?

192.168.202.79 with 2,620 unique domains.

6. Most common query types?

A (212,477), NB (77,199), AAAA (54,519), PTR (53,017), TXT (12,870), SRV (12,268). Uncommon: AXFR (440), HINFO (30), NAPTR (27).

7. Response codes and frequency?

`-` 372,201; NXDOMAIN 50,330; NOERROR 2,665; REFUSED 1,797; NXRRSet 728; SERVFAIL 105; NOTIMP 44; NOTAUTH 32; FORMERR 14; plus rare codes (`unassigned-*`, YXRRSET, YXDOMAIN, NOTZONE).

8. Which clients generated the most NXDOMAIN?

192.168.202.103 (7,471), 192.168.202.138 (4,762), 192.168.202.79 (4,711), 192.168.202.83 (4,453) and 192.168.202.140 (3,835). `.79`, `.138` and `.140` failed on more than 2,000 different names (scanning pattern); `.103` and `.83` failed on 62 and 18 names (retries).

9. Which domains have the most distinct source IPs?

`safebrowsing.clients.google.com` (59 clients), `creativecommons.org` (47), and `dokuwiki.org`, `jigsaw.w3.org`, `validator.w3.org`, `www.php.net` (46 each). The `-` entry (127) is not a domain.

10. Are there noticeable spikes? When?

Yes. **16 Mar 16:00 to 19:00 UTC**, peaking at 73,300 events at 18:00 against a normal 8,000 to 20,000 per hour. It is driven by 10.10.117.210 (up to 6,157 queries in 5 minutes at 17:05 UTC). Smaller bursts on 17 Mar come from 192.168.202.79 and .138.

11. Five rare domains and why each is worth reviewing

(1) `1.229.168.192.bl.spamcop.net` and (2) `1.229.168.192.list.quorum.to`: blocklist checks of an internal address, typical of scanning tools; (3) `metasploitable` and (4) `acme-texmcxhenx`: probe-style names seen once and unresolved; (5) `0zqsnofz`: random single-label name from 192.168.202.140. Also `100.204.14.72.in-addr.arpa`: a single reverse lookup that could be part of an address sweep. Rarity alone is not proof of malice.

12. Three client/domain relationships to investigate

(1) **192.168.202.79 to `*.hec.net`: **437 zone-transfer requests to 10 DNS servers, 415 answered with internal host data. (2)** 192.168.202.140 / .79 to `rootshell-security.net`: **requested by seven clients, all NXDOMAIN. (3)** 192.168.202.108 / .79 to blocklist domains**: 240 and 84 DNSBL events, unusual for normal users.

13. Security-sensitive keywords?

The keyword search matched 9,600 events, almost all legitimate (Symantec LiveUpdate, Windows Update, `login.live.com`, `login.oscar.aol.com`). **`secure.informaction.com`** (874 queries, 29 clients) needs validation. No phishing-style names were found.

14. Domain requiring threat-intelligence validation

`rootshell-security.net`. VirusTotal (4 Oct 2026): 4 of 91 vendors flagged it. URLhaus (4 Oct 2026): no entry. Result: mixed and inconclusive. The domain is two years old but the log is from 2012, and all queries were NXDOMAIN (Section 6).

15. Anomaly with a benign explanation

10.10.117.210 is the top client (75,943 queries) but queried only 250 domains, mainly `teredo.ipv6.microsoft.com` (27,425), Google, Norton/Symantec and update services, with no failures. It is probably a gateway/proxy or a host running automated software. **Evidence needed:** DHCP and asset records, how many users sit behind the address, and DNS-server logs. A second example is 192.168.202.103/.83: thousands of NXDOMAIN for only 62 and 18 names, which indicates software retries.

16. Incident hypothesis (evidence versus assumptions)

Evidence: 192.168.202.79 sent 437 AXFR requests for 7 zones to 10 DNS servers (415 answered NOERROR with internal records), about 3,700 PTR lookups across nine subnets, queried 2,620 unique domains (2,557 unique failures) and made blocklist lookups. **Assumptions (not proven):** the host performed reconnaissance of `hec.net`, either unauthorised or as an approved scan, and complete zones were transferred. **Conclusion:** suspicious and requires validation.

17. Additional logs to request

Firewall/NetFlow/Zeek `conn.log` (TCP/53 volume between .79 and the DNS servers); DNS-server logs (which zone transfers were allowed); DHCP and asset inventory (owner of .79); EDR/endpoint process logs from that host; the vulnerability-scan and penetration-test schedule; authentication logs.

9. Final Investigation: SOC Analyst Assessment

9.1 Anomaly (one sentence)

Internal host 192.168.202.79 sent 437 zone-transfer (AXFR) requests to the DNS servers, swept many subnets with reverse lookups and caused thousands of failed lookups, which looks like network reconnaissance of the `hec.net` environment.

9.2 SPL that found it

SPL 21 (see Figure 22)

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search query_type=AXFR | stats count dc(fqdn)
as zones dc(dest_ip) as dns_servers min(_time) as first max(_time) as last by src_ip |
convert ctime(first) ctime(last)
```

The host was first flagged by the unique-domains search (SPL 16), where it queried 2,620 unique domains, the most of any client.

9.3 Affected systems and time period

Item	Detail
Client	192.168.202.79 (related leads: .102, .108, .115, .138, .140)
Targets	10 DNS servers, mainly `192.168.2N.25` (N = 21 to 28); 7 zones: `ma`, `fl`, `wa`, `ny`, `ca`, `ky`, `tx.hec.net`
Domain of interest	`rootshell-security.net` (7 clients, all NXDOMAIN)
Time period (UTC)	AXFR: 16 Mar 2012 15:57 to 17 Mar 2012 15:50. Host activity: 16 Mar 12:30 to 17 Mar 20:38.

9.4 Pivots

Pivot	Finding
Query type	437 AXFR; 3,703 PTR
Response code	415 AXFR answered NOERROR with internal hostnames and IPs; 4 NOTAUTH; 2 REFUSED; 2,702 PTR answered NXDOMAIN
Frequency and diversity	2,620 unique domains; 2,557 unique failed names; burst of 2,126 queries in 5 minutes on 17 Mar
Related clients	Blocklist lookups from .79 and .108; `version.bind` from .102 and .115; failed-name diversity on .138 and .140

9.5 Threat-intelligence validation

`rootshell-security.net` (requested by 192.168.202.79 and six other clients), checked 4 Oct 2026: VirusTotal flags 4 of 91 vendors; URLhaus has no entry. The result is mixed and inconclusive (Section 6). The case rests on the host's internal behaviour.

9.6 Conclusion

> **Outcome: Suspicious and requires validation**
>
> **Why:** one host requested zone transfers for 7 zones from 10 DNS servers, received answers containing internal host data, swept nine subnets and produced more than 2,500 unique failed lookups. No other host shows this combination.
>
> **Not proven:** intent, and whether complete zones were copied. An approved security scan would look the same, so the activity is not called malicious.

9.7 Next three actions as an L1 SOC analyst

1. **Identify the owner and check authorisation.** Use DHCP and asset records to find who owns 192.168.202.79, then ask IT or change management whether a scan or penetration test was approved for 16 to 17 March.
2. **Confirm what was transferred.** Review firewall, NetFlow or `conn.log` data for TCP/53 between 192.168.202.79 and the DNS servers, and DNS-server logs for allowed zone transfers.
3. **Escalate and contain if unapproved.** Raise a ticket to L2, review the host with EDR, and recommend restricting AXFR on the DNS servers to approved secondary servers only. Review 192.168.202.102, .108, .115, .138 and .140 for similar behaviour.

10. Conclusion

Analysis of 427,935 DNS events found that most activity is normal background traffic: Windows IPv6 tunnelling, time synchronisation, browser safe-browsing checks and software updates. The busiest client (10.10.117.210) repeats a small set of domains and is assessed as expected behaviour.

The most significant anomaly is host **192.168.202.79**. It sent 437 of the 440 zone-transfer requests across seven zones and ten DNS servers (415 answered with internal host records), swept nine subnets with about 3,700 reverse lookups, queried 2,620 unique domains with 2,557 unique failures and performed blocklist lookups. Supporting observations include `version.bind` fingerprinting from 192.168.202.102 and .115, blocklist lookups from 192.168.202.108, and a shared unresolved domain, `rootshell-security.net`, whose threat-intelligence result is mixed.

Outcome: suspicious and requires validation. The behaviour matches reconnaissance, but the dataset cannot show intent, the tool used or the data volume, so the activity is not classified as malicious.

10.1 Limitations

- The dataset contains DNS only: no connection volumes, user identities, process data or DNS-server logs.
- The log dates from March 2012, while threat-intelligence results reflect October 2026.
- Splunk displays times in IST; the data is in UTC (conversion stated throughout).
- 372,201 events show `-`, meaning no reply was recorded, which limits response analysis.
- A significant gap with no events (about 23:00 UTC 16 Mar to 12:00 UTC 17 Mar) means the capture is not continuous.
- NOERROR on AXFR and the answers returned strongly suggest transfers were allowed, but they do not prove complete zones were copied.

11. Recommended Next Steps

1. **Identify the owner of 192.168.202.79** from DHCP and asset records and confirm with IT or change management whether a scan or penetration test was approved for 16 to 17 March 2012.
2. **Confirm what was transferred** by reviewing firewall, NetFlow or `conn.log` data for TCP/53 traffic between 192.168.202.79 and the DNS servers, and the DNS-server logs.
3. **Escalate to L2 if the activity was not approved**, and review the host with EDR or endpoint logs.
4. **Harden the DNS servers** by restricting AXFR to approved secondary servers only.
5. **Review related hosts** (192.168.202.102, .108, .115, .138 and .140) for similar scanning behaviour.
6. **Monitor `rootshell-security.net`** and re-check its reputation, since the current result is mixed.
7. **Create a detection** for AXFR requests from non-DNS hosts and for clients with a high ratio of unique failed names.

Appendix A: SPL Searches Used

All searches use index `dns_lab`, sourcetype `dns_sample` and time range **All time**. The macro `dnsf` extracts the DNS fields (Appendix B). The type `-` in the Zeek log stands for "no value".

SPL 1: Validation

```
index=dns_lab sourcetype=dns_sample | head 20
```

SPL 2: Field summary

```
index=dns_lab sourcetype=dns_sample | `dnsf` | fieldsummary | sort - count
```

SPL 3: DNS fields only

```
index=dns_lab sourcetype=dns_sample | `dnsf` | fieldsummary | search field IN
("src_ip","dest_ip","fqdn","query_type","response_code","src_port","dest_port","proto","uid"
,"ts") | table field count distinct_count | sort - count
```

SPL 4: Task 01, event count and time range

```
index=dns_lab sourcetype=dns_sample | stats count min(_time) as first max(_time) as last |
convert ctime(first) ctime(last)
```

SPL 5: Task 02, top domains

```
index=dns_lab sourcetype=dns_sample | `dnsf` | top limit=20 fqdn
```

SPL 6: Task 03, top clients

```
index=dns_lab sourcetype=dns_sample | `dnsf` | top limit=20 src_ip
```

SPL 7: Profile of 10.10.117.210

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=10.10.117.210 | stats count
dc(fqdn) as unique_domains min(_time) as first max(_time) as last | convert ctime(first)
ctime(last)
```

SPL 8: Domains/types/responses of 10.10.117.210

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=10.10.117.210 | stats count by
fqdn query_type response_code | sort - count | head 15
```

SPL 9: Task 04, query types

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by query_type | sort - count
```

SPL 10: Task 05, response codes

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by response_code | sort - count
```

SPL 11: Task 06, activity over time

```
index=dns_lab sourcetype=dns_sample | timechart span=1h count
```

SPL 12: Task 07, client/domain pairs

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count by src_ip fqdn | sort - count |
head 30
```

SPL 13: Task 08, rare domains

```
index=dns_lab sourcetype=dns_sample | `dnsf` | rare limit=30 fqdn
```

SPL 14: Task 09, NXDOMAIN pairs

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search response_code=NXDOMAIN | stats count
by src_ip fqdn | sort - count | head 30
```

SPL 15: Task 10, keyword hunt

```
index=dns_lab sourcetype=dns_sample | regex _raw="(?i)(login|verify|secure|update|account|
password|wallet)" | `dnsf` | stats count by fqdn src_ip | sort - count
```

SPL 16: Phase 3A, volume vs unique domains

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats count as dns_queries dc(fqdn) as
unique_domains by src_ip | sort - dns_queries | head 20
```

SPL 17: Phase 3B, domains by distinct clients

```
index=dns_lab sourcetype=dns_sample | `dnsf` | stats dc(src_ip) as unique_clients count as
queries by fqdn | sort - unique_clients | head 30
```

SPL 18: Phase 3C, NXDOMAIN by client

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search response_code=NXDOMAIN | stats count
as nxdomain_count dc(fqdn) as unique_failed_domains by src_ip | sort - nxdomain_count
```

SPL 19: Phase 3D, query bursts

```
index=dns_lab sourcetype=dns_sample | `dnsf` | bin _time span=5m | stats count as queries by
_time src_ip | sort - queries | head 50
```

SPL 20: Phase 3E, domain pivot

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn="rootshell-security.net" | stats
count values(src_ip) as clients values(query_type) as query_types by fqdn
```

SPL 21: AXFR requests by client

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search query_type=AXFR | stats count dc(fqdn)
as zones dc(dest_ip) as dns_servers min(_time) as first max(_time) as last by src_ip |
convert ctime(first) ctime(last)
```

SPL 22: AXFR zones and servers

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR
| stats count by fqdn dest_ip | sort - count
```

SPL 23: AXFR responses with answers

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR
response_code=NOERROR | rex field=_raw "\t(?<answers>[^\t]+)\t(?<TTLs>[^\t]+)\t(?
<rejected>[^\t]+)$" | table _time fqdn dest_ip answers | head 10
```

SPL 24: Raw AXFR events

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=AXFR
| head 3 | table _time _raw
```

SPL 25: PTR sweep by subnet

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 query_type=PTR |
rex field=fqdn "^\d+\.(?<third_octet>\d+)\.168\.192\.in-addr\.arpa" | stats count by
third_octet | sort - count
```

SPL 26: Query types/responses for 192.168.202.79

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search src_ip=192.168.202.79 | stats count by
query_type response_code | sort - count
```

SPL 27: version.bind queries

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn="version.bind" | stats count
dc(dest_ip) as dns_servers by src_ip | sort - count
```

SPL 28: DNSBL lookups

```
index=dns_lab sourcetype=dns_sample | `dnsf` | search fqdn IN
("*sorbs.net","*spamhaus.org","*spamcop.net","*spamrats.com","*quorum.to","*tornevall.org","
*apews.org","*nszones.com") | stats count dc(fqdn) as lookups by src_ip | sort - count
```

Appendix B: Field-Extraction Macro `dnsf`

Created in **Settings → Advanced search → Search macros** (name `dnsf`, no leading pipe). The log has no header row, so the fields are named according to the Zeek DNS log layout. The fields after `response_code` (AA, TC, RD, RA, Z, answers, TTLs, rejected) are not extracted by the macro; `answers` is extracted inside SPL 23 where needed.

```
rex "^(?<ts>[^\t]+)\t(?<uid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<src_port>[^\t]+)\t(?<dest_ip>[^\
t]+)\t(?<dest_port>[^\t]+)\t(?<proto>[^\t]+)\t(?<trans_id>[^\t]+)\t(?<fqdn>[^\t]+)\t(?
<qclass>[^\t]+)\t(?<qclass_name>[^\t]+)\t(?<qtype>[^\t]+)\t(?<query_type>[^\t]+)\t(?
<rcode>[^\t]+)\t(?<response_code>[^\t]+)"
```

Time conversion reference

Splunk displays (IST)	UTC	Example
03/16/2012 18:00:05	16 Mar 2012 12:30:05	First event
03/17/2012 21:20:28	17 Mar 2012 15:50:28	Last AXFR request
03/18/2012 02:29:51	17 Mar 2012 20:59:51	Last event